

Dynamical Mean Field Theory (DMFT) and DFT+DMFT

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Dynamical Mean Field Theory

DFT: an electron in the **effective field** of other electrons

DMFT: a correlated atom in the **effective field** of other atoms

Dynamical Mean Field Theory

DFT: an electron in the **effective field** of other electrons

DMFT: a correlated atom in the **effective field** of other atoms

- **Dynamical :**

- The effective field is frequency dependent = energy levels generating the field are distributed in energy.
- In contrast to static mean field theory of ferromagnetism in the Ising model.
- The frequency dependence of this field is not created by interactions but is required to treat the correlated atom.

Dynamical Mean Field Theory

DFT: an electron in the **effective field** of other electrons

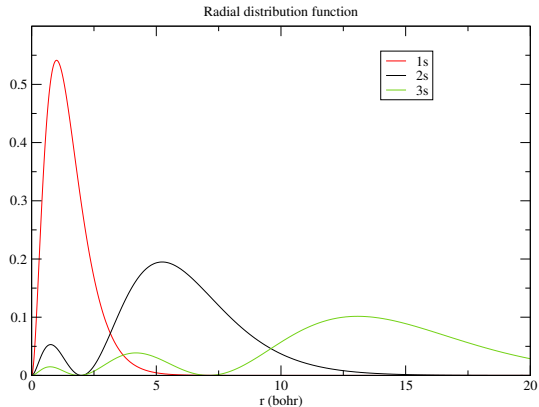
DMFT: a correlated atom in the **effective field** of other atoms

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- **Dynamical :** the self-energy is frequency dependent, in contrast to DFT+ U and DFT, allowing to describe strong correlations and its features in photoemission.

Atomic orbitals 1s, 2s and 3s



$$\Psi_{nlm}(\mathbf{r}) = R_{nl}(r)Y_{lm}(\theta, \phi)$$

Radial distribution function is $(rR_{nl}(r))^2$

5s/4d/5p

Lanthanides 4fActinides 5f

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THE PERIODIC TABLE

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37 Rb	38 Sr	39 Y	40 Zr	41 Nb	42 Mo	43 Tc	44 Ru	45 Rh	46 Pd	47 Ag	48 Cd	49 In	50 Sn	51 Sb	52 Te	53 I	54 Xe
55 Cs	56 Ba	57-71 La-Lu	72 Hf	73 Ta	74 W	75 Re	76 Os	77 Ir	78 Pt	79 Au	80 Hg	81 Tl	82 Pb	83 Bi	84 Po	85 At	86 Rn
87 Fr	88 Ra	89-103 Ac-Lr	104 Rf	105 Db	106 Sg	107 Bh	108 Hs	109 Mt	110 Ds	111 Rg	112 Cn	113 Nh	114 Fl	115 Mc	116 Lv	117 Ts	118 Og

4s/3d/4p

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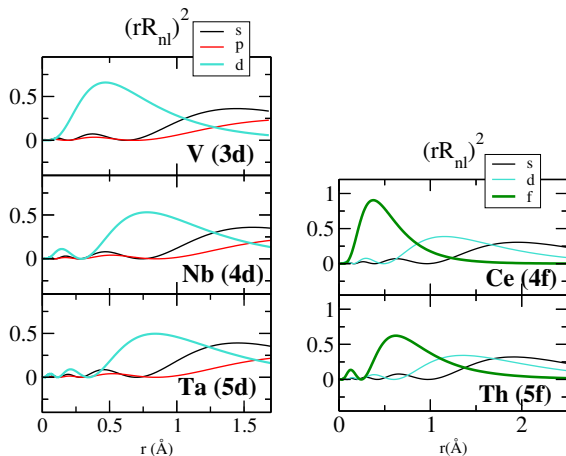
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Localization of $3d$, $4f$ and $5f$ orbitals.

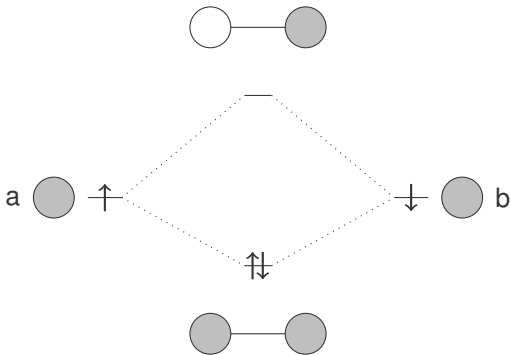


- $\Psi_{nlm}(\mathbf{r}) = R_{nl}(r)Y_{lm}(\theta, \phi)$
- For $1s, 2p, 3d, 4f$, R has no node, their maxima are thus closer to the nucleus
- $3d$ and $4f$ orbitals are more localized.

Hydrogen Molecule

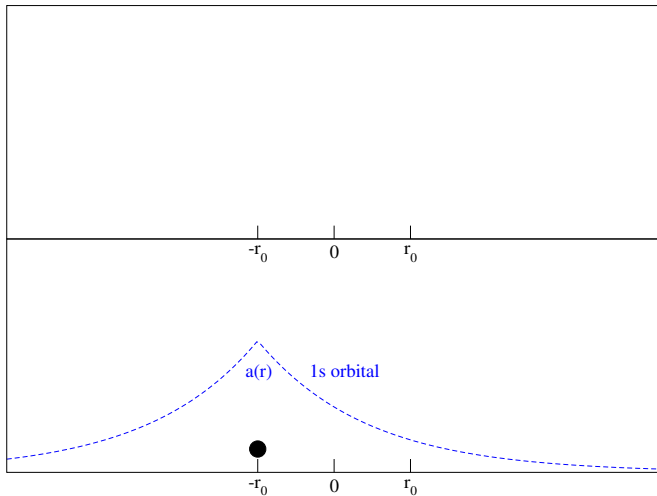
Non diagonal elements of the Hamiltonian creates chemical bonding

antibonding state $\sigma_u = \frac{a-b}{\sqrt{2}}$

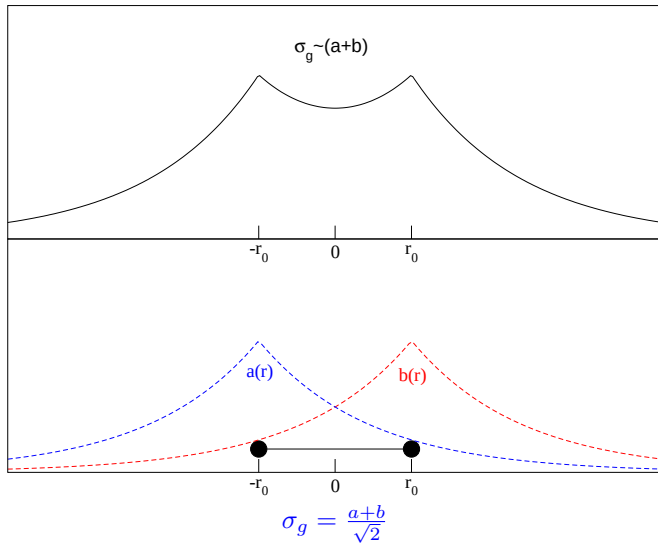


bonding state $\sigma_g = \frac{a+b}{\sqrt{2}}$

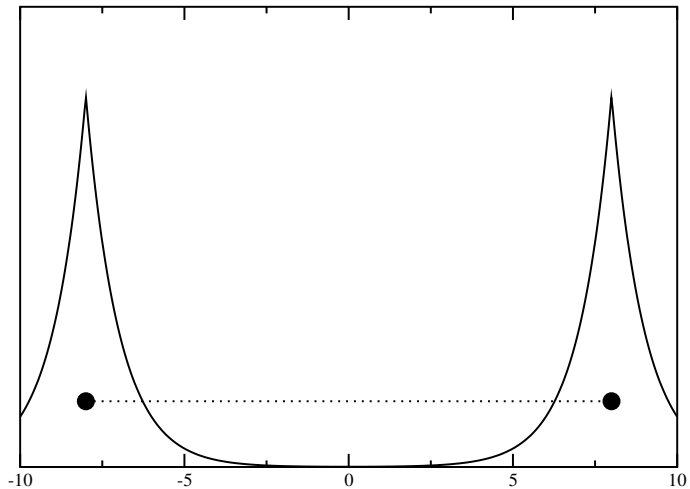
Hydrogen atom: 1s orbital



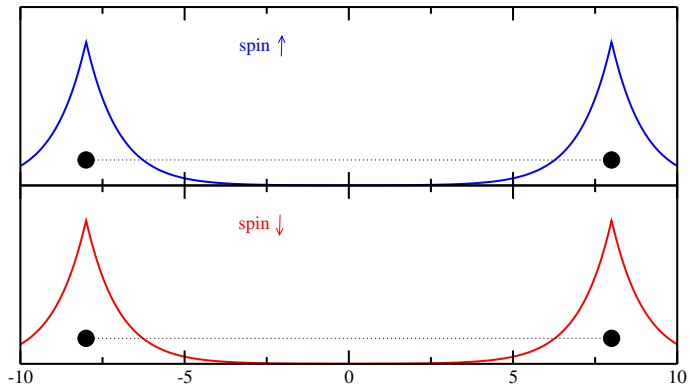
Hydrogen molecule: molecular orbital.



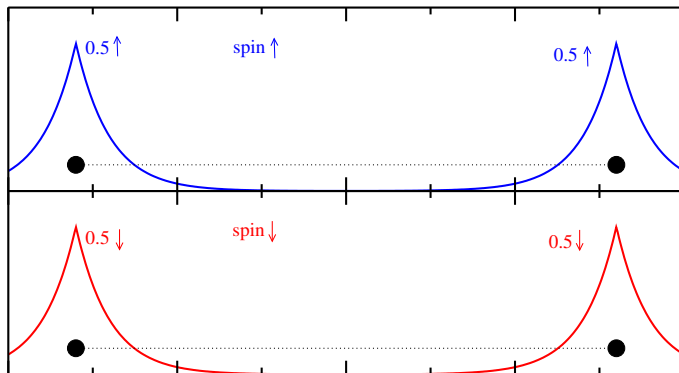
Hydrogen molecule: stretched.



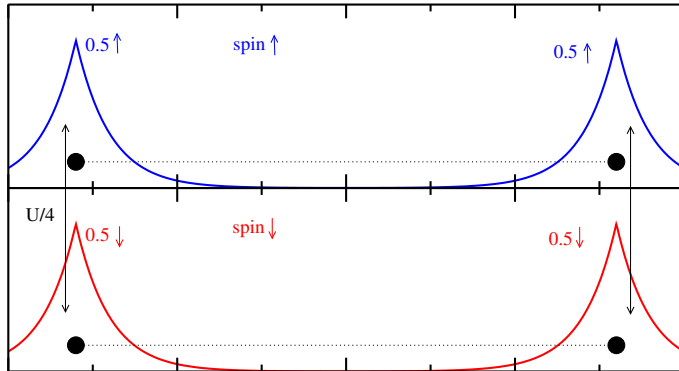
Hydrogen molecule: same orbital for both spins !



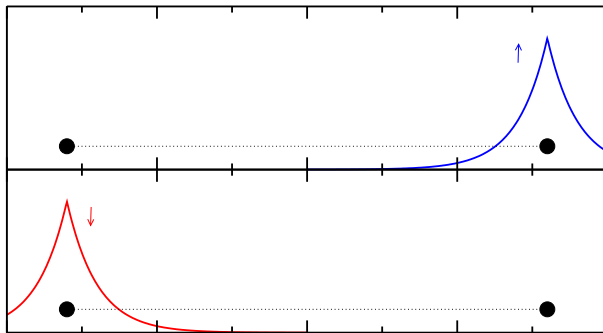
Hydrogen molecule: same orbital for both spins !



Hydrogen molecule: dissociation limit is bad



A solution: break symmetry

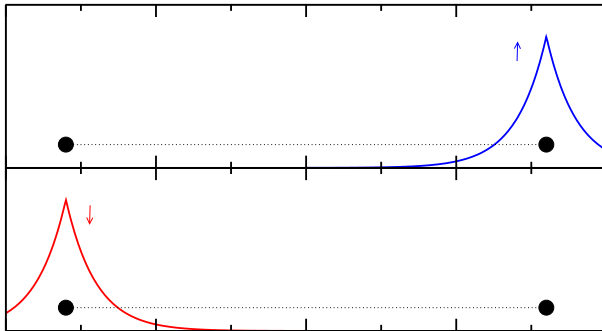


no more artificial delocalization, no more interaction

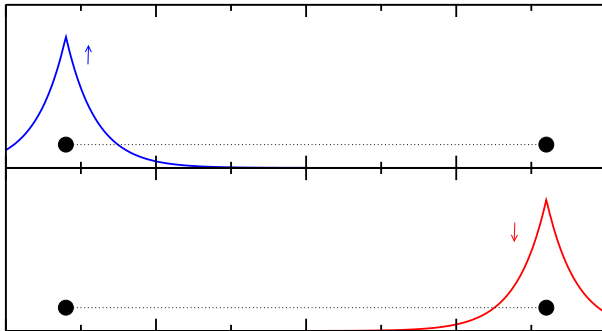
But

- Symmetry breaking. An artificial magnetism is induced.
- A static theory, which overestimates localization.

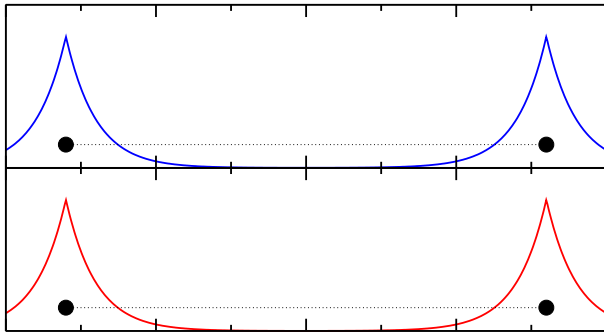
Solution 1: localization: lowers interaction



Solution 2: localization: lowers interaction



Solution 3: delocalization: lowers kinetic energy

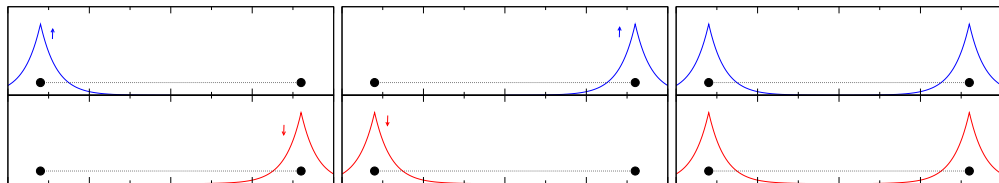


The exact solution

$$|\Psi\rangle = c_1|1 : \text{localized } \uparrow\downarrow\rangle$$

$$+ c_2|2 : \text{localized } \downarrow\uparrow\rangle$$

$$+ c_3|3 : \text{delocalized}\rangle$$

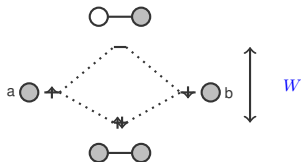


• This mixing of configuration correctly describes the system (magnetism, structural properties).

- $c_1 = c_2$: no ordered magnetism.
- c_3 increases if distance between atoms lowers.

• If $W \gg U$ then $c_1 = c_2 \ll c_3$

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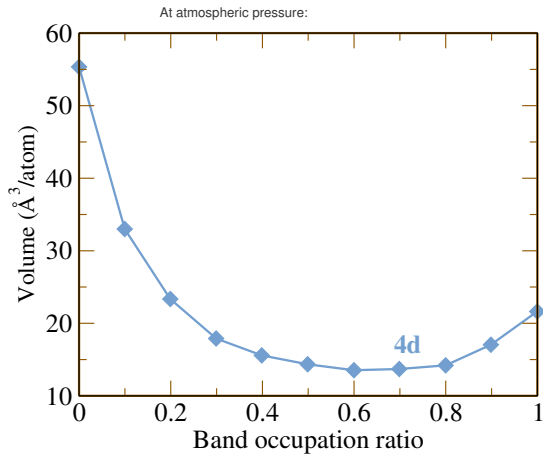
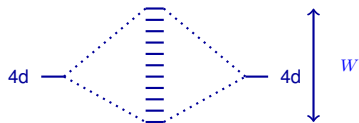
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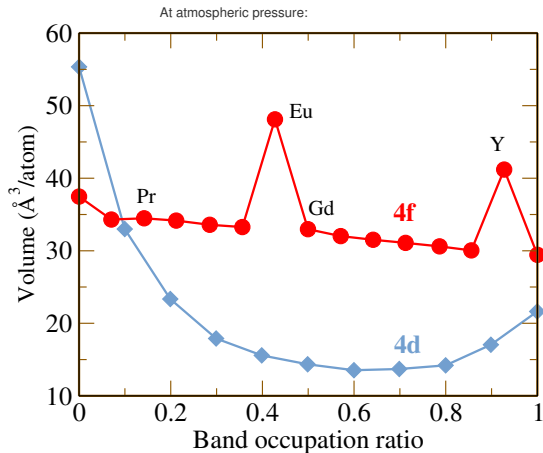
Localization in f electrons systems



[Mac Mahan, et al J. Comp.-Aid. Mater. Des. 5, 131 (1998)]

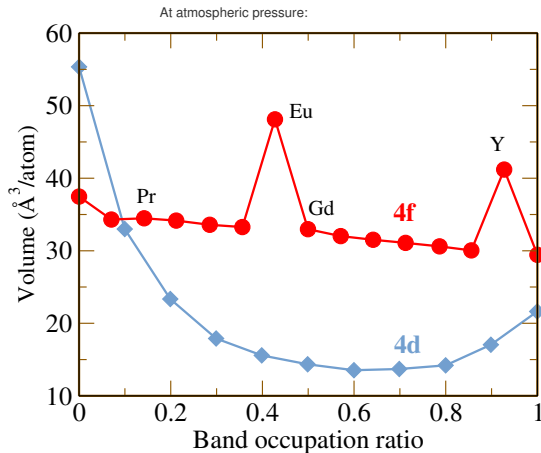
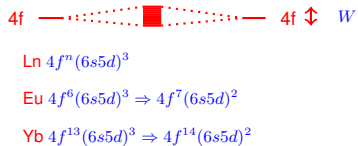
Localization in f electrons systems

4f —·····■·····— 4f $\uparrow\downarrow$ w



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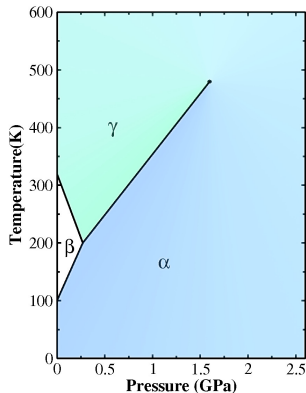
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Isostructural transition in Cerium

Isostructural transition $\frac{V_\gamma - V_\alpha}{V_\gamma} = 15\%$, ends at a critical point

Electronic configuration $4f^1$.

- α phase: Pauli paramagnetism
 $\Rightarrow \alpha$ phase: f e^- more delocalized.
- γ phase: Curie Paramagnetism
 $\Rightarrow \gamma$ phase: f e^- is localized



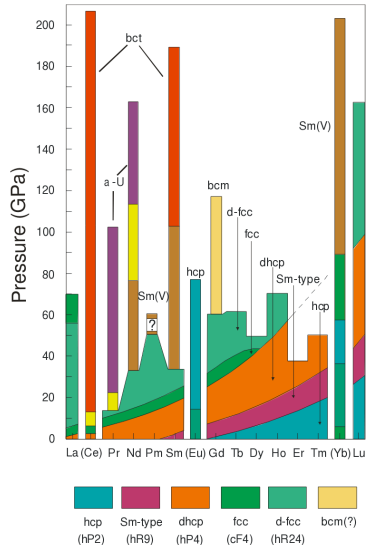
[Johansson, B. Phil. Mag. **30**, 469 (1974)]

Transition in lanthanides.

We now discuss the equation of states of lanthanides as a function of pressure.

- At low pressure, compact structures.
- Under pressure, more distorted structure
 - f electrons participate to the bonding

Figure from [Schiwek, (2002)]



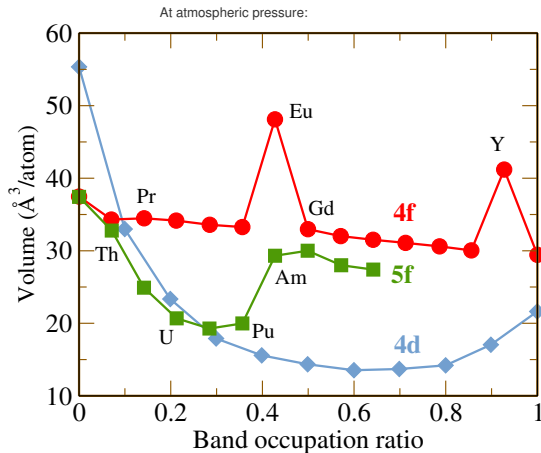
Localization in f electrons systems

$4d$ element: filling of the $4d$ band
(Bonding states and antibonding):

$4d$ electrons are **delocalized**.

Lanthanides: $4f$ electrons **localized**,
negligible overlap between $4f$ orbitals.

Actinides: intermediate case of localization.



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THE EXACT HAMILTONIAN

The exact hamiltonien is (i, j are electrons)

$$H = \sum_{i=1}^N \left[-\frac{1}{2} \nabla_{\mathbf{r}_i}^2 + V_{\text{ext}}(\mathbf{r}_i) \right] + \frac{1}{2} \sum_{i \neq j} \frac{1}{|\mathbf{r}_i - \mathbf{r}_j|}$$

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- One orbital per atom: i and j indicates thus only the atom on which orbitals are centered.
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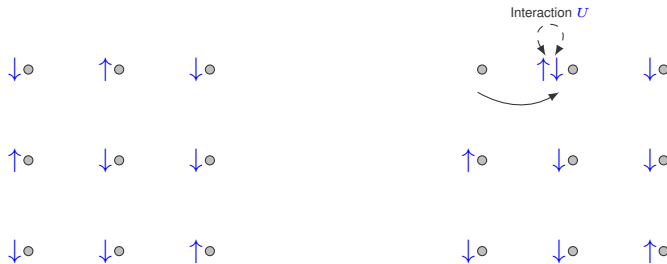
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$$H = \sum_i \epsilon_0 \left(\underbrace{n_{i\uparrow} + n_{i\downarrow}}_{c_{i\uparrow}^\dagger c_{i\uparrow}} \right) + \underbrace{\sum_{j \neq i, \sigma = \uparrow, \downarrow} t_{ij} c_{i\sigma}^\dagger c_{j\sigma}}_{\text{one electron term : delocalization}} + \underbrace{\sum_i U \hat{n}_{i\uparrow} \hat{n}_{i\downarrow}}_{\text{interaction term : localization}}$$

The Hubbard model: Competition between localization and delocalization

$$H = \sum_i \epsilon_0 (n_{i\uparrow} + n_{i\downarrow}) + \sum_{j \neq i, \sigma = \uparrow, \downarrow} t_{ij} c_{i\sigma}^\dagger c_{j\sigma} + \sum_i U \hat{n}_{i\uparrow} \hat{n}_{i\downarrow}$$



U is the energy repulsion of two electrons on the same site.

- For large value of the interaction U , electrons are localized
- For low value of the interaction U , electrons are delocalized

The Hubbard model $U = 0$

In this limit, we neglect the interaction term.

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In this case, one can easily solve this non interacting Hamiltonian either by direct diagonalisation or by using Bloch states:

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$$|\psi_k\rangle = \frac{1}{\sqrt{N}} \sum_i |\phi_{T_i}\rangle e^{ikT_i}$$

where $|\phi_{T_i}\rangle$ are atomic orbitals: $\langle r | \phi_{T_i} \rangle = \phi(r - T_i)$ on site i . Does it satisfy the Bloch theorem ?

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$$\epsilon_k = \epsilon_0 + \frac{1}{N} \sum_{i \neq j} t_{ij} e^{-ik(T_i - T_j)} = \epsilon_0 + \frac{1}{N} \sum_{i=1}^N [t_{i,i+1} e^{-ik(-a)} + t_{i,i-1} e^{-ik(a)}]$$

The Hubbard model $U = 0$

We also have the change of basis for creation and annihilation operators:

$$c_i^\dagger = \frac{1}{\sqrt{N}} \sum_k c_k^\dagger e^{-ikT_i} \quad c_i = \frac{1}{\sqrt{N}} \sum_k c_k e^{ikT_i}$$

As a consequence, one can show that (using that $\frac{1}{N} \sum_i e^{i(k-k')T_i} = \delta_{kk'}$, show it!).

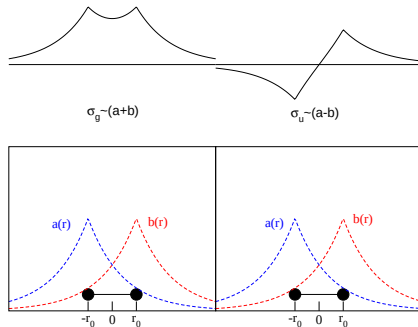
$$H = \sum_i \epsilon_0 (n_{i\uparrow} + n_{i\downarrow}) + \sum_{j \neq i, \sigma = \uparrow, \downarrow} t_{ij} c_{i\sigma}^\dagger c_{j\sigma} = \sum_k \epsilon_k c_k^\dagger c_k \quad \text{with} \quad \epsilon_k = \epsilon_0 + \frac{1}{N} \sum_{i \neq j} t_{ij} e^{-ik(T_i - T_j)}$$

In a simple case (one dimension and if t_{ij} is non zero only for neighboring atoms), we have :

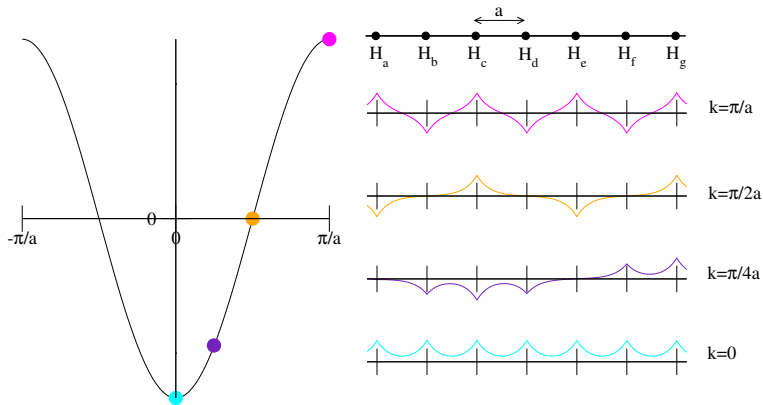
$$\epsilon_k = \epsilon_0 + \frac{1}{N} \sum_{i \neq j} t_{ij} e^{-ik(T_i - T_j)} = \epsilon_0 + \frac{1}{N} \sum_{i=1}^N [t_{i,i+1} e^{-ik(-a)} + t_{i,i-1} e^{-ik(a)}] = \epsilon_0 + 2t \cos(ka)$$

Toward the 1D chain: H_2

- For hydrogen molecule, hopping t induces a bonding state and an antibonding states for the molecule.



The chain of atoms: $\epsilon_k = \epsilon_0 + 2t\cos(ka)$ $t < 0$



$$\psi_k(\mathbf{r}) = \sum_i e^{i\mathbf{k}T_i} \phi(\mathbf{r} - T_i) = e^{i\mathbf{k}(\mathbf{r})} \sum_n e^{-i\mathbf{k}(\mathbf{r}-T_i)} \phi(\mathbf{r} - T_i) = e^{i\mathbf{k}(\mathbf{r})} u_{\mathbf{k}}(\mathbf{r}) \quad [T_i = ia]$$

Hubbard model with $t = 0$

$$H = \sum_i \epsilon_0 (n_{i\uparrow} + n_{i\downarrow}) + \sum_{j \neq i, \sigma = \uparrow, \downarrow} t_{ij} c_{i\sigma}^\dagger c_{j\sigma} + \sum_i U \hat{n}_{i\uparrow} \hat{n}_{i\downarrow}$$

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If $t = 0$ then

$$H = \sum_i \epsilon_0 (n_{i\uparrow} + n_{i\downarrow}) + \sum_i U \hat{n}_{i\uparrow} \hat{n}_{i\downarrow}$$

Atoms are disconnected, one can study one single atom !

$$H = \sum_i [\epsilon_0 (n_{i\uparrow} + n_{i\downarrow}) + U \hat{n}_{i\uparrow} \hat{n}_{i\downarrow}] = \sum_i H_{\text{atomic}}^i$$

Hubbard model with $t = 0$

The isolated atom limit: $t = 0$

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$$H = U n_{\uparrow} n_{\downarrow} + \epsilon_0 (n_{\uparrow} + n_{\downarrow})$$

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How many electrons can we put in this system ?

Hubbard model with $t = 0$

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$$H = U n_{\uparrow} n_{\downarrow} + \epsilon_0 (n_{\uparrow} + n_{\downarrow})$$

How many electrons can we put in this system ?

What is the size of the Hilbert space ?

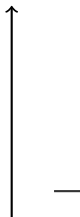
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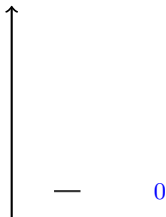
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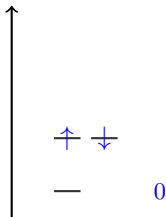
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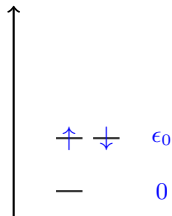
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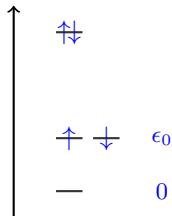
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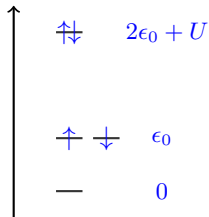
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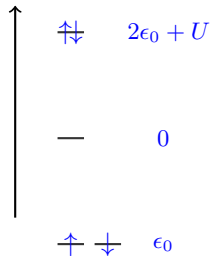
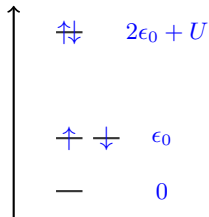
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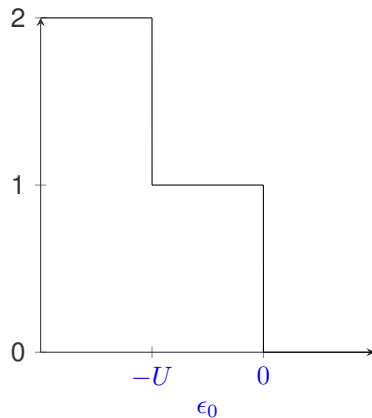
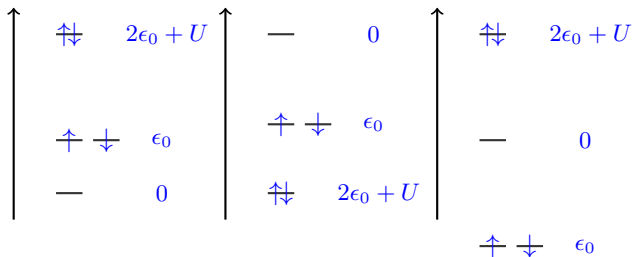
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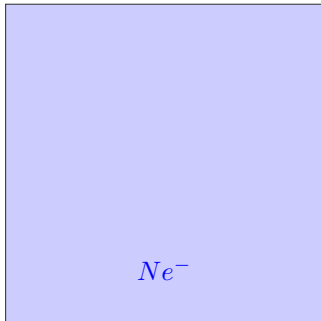
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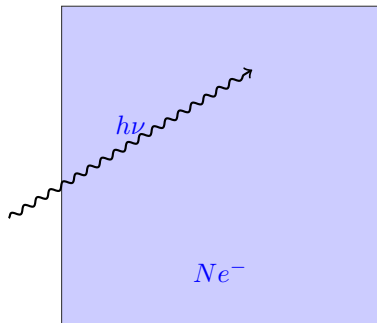
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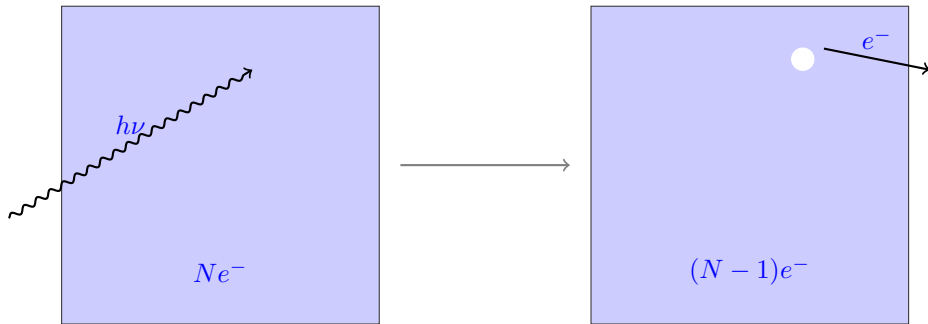
Direct photoemission



Direct photoemission



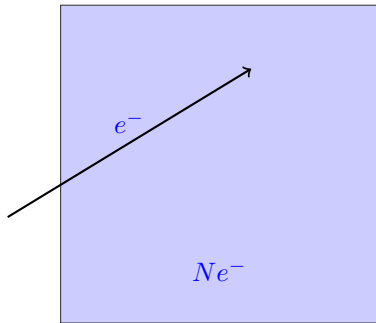
Direct photoemission



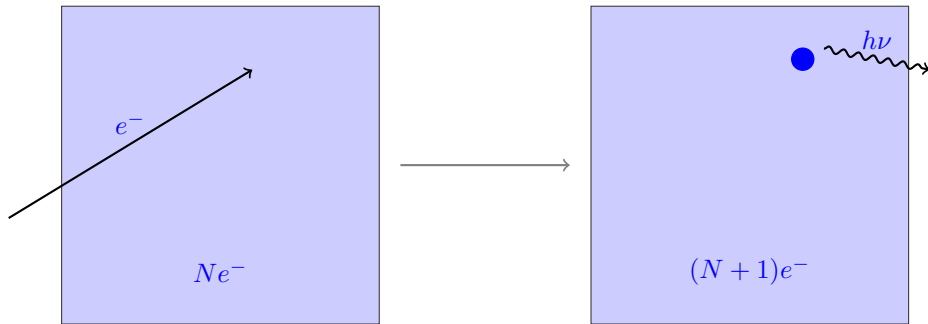
$$h\nu + E_N = E_{\text{kin}} + E_{N-1}$$

One can measure $E_N - E_{N-1}$

Inverse photoemission



Inverse photoemission



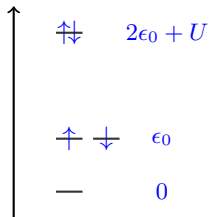
$$E_{\text{kin}} + E_N = h\nu + E_{N+1}$$

One can measure $E_{N+1} - E_N$

Hubbard bands

The isolated atom limit: $t = 0$

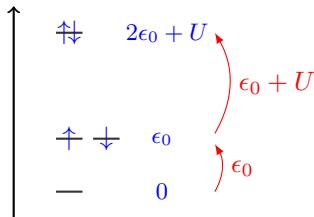
$$H = U n_{\uparrow} n_{\downarrow} + \epsilon_0 (n_{\uparrow} + n_{\downarrow}) = U n_{\uparrow} n_{\downarrow} + \epsilon_0 (n_{\uparrow} + n_{\downarrow})$$



Hubbard bands

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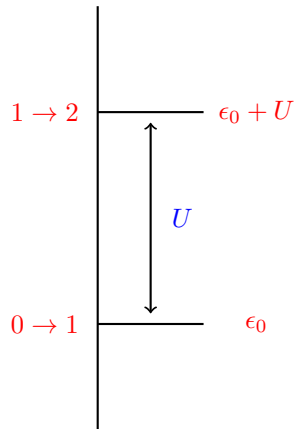
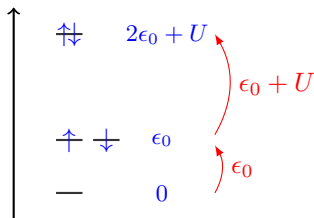
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Hubbard bands

The isolated atom limit: $t = 0$

$$H = Un_{\uparrow}n_{\downarrow} + \epsilon_0(n_{\uparrow} + n_{\downarrow}) = Un_{\uparrow}n_{\downarrow} + \epsilon_0(n_{\uparrow} + n_{\downarrow})$$



Atomic case: Green's function

- We study the atomic problem with 1 electron.
- Green's function of the atom and Self-energy. Let's use $\epsilon_0 = -U/2$ and the Lehman representation of the

$$iG(\mathbf{r}, \mathbf{r}', t-t') = \langle N, 0 | T [\Psi(\mathbf{r}t) \Psi^\dagger(\mathbf{r}'t')] | N, 0 \rangle$$

Closure relation

$$\sum_{M,i} |M,i\rangle \langle M,i|$$

Lehman representation:

$$G(\mathbf{r}, \mathbf{r}', \omega) = \sum_i \frac{f_i(\mathbf{r}) f_i^*(\mathbf{r}')}{\omega - \epsilon_i \pm i\eta}$$

Green's function :

$$G(\omega) = \frac{1/2}{\omega + U/2} + \frac{1/2}{\omega - U/2} = \frac{\omega}{\omega^2 - (U/2)^2}$$

Equation of motion for the non interacting Green's function.

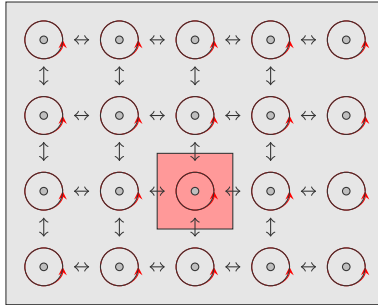
$$(\omega - \epsilon_0) \mathcal{G}_0(\omega) = 1 \Rightarrow \mathcal{G}_0(\omega) = \frac{1}{\omega - \epsilon_0} = \frac{1}{\omega + U/2}$$

- The Dyson eq writes:

$$\Sigma(\omega) = \mathcal{G}_0^{-1} - G^{-1} = \omega + U/2 - \frac{\omega^2 - (U/2)^2}{\omega} = U/2 + \frac{(U/2)^2}{\omega}$$

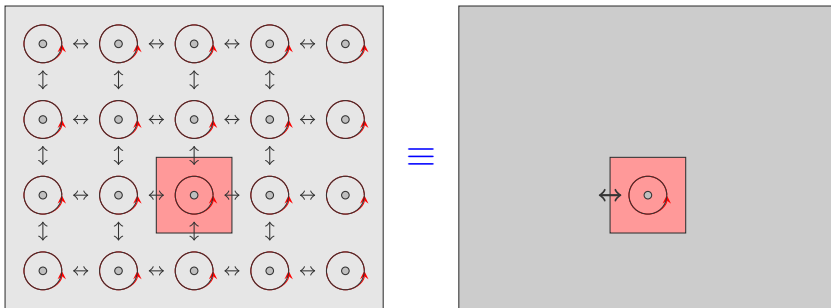
Dynamical Mean Field Theory

The Hubbard model physics can be mimicked by an Anderson model + Self-consistency



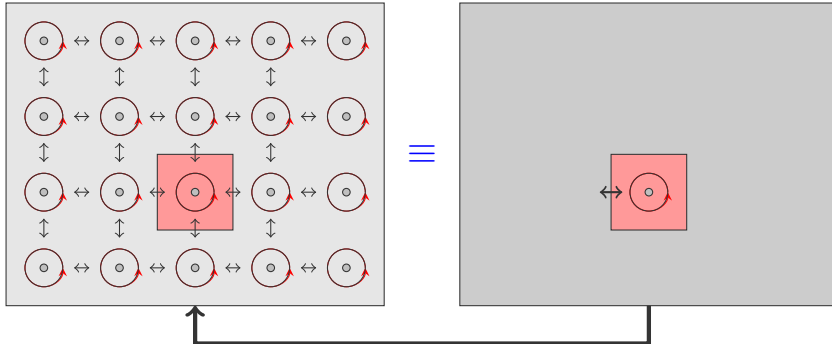
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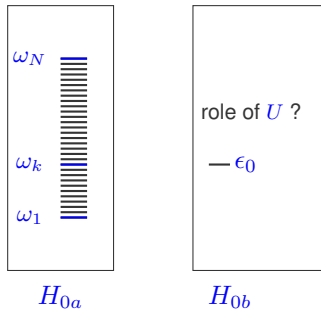


W. Metzner and D. Vollhardt Phys. Rev. Lett. 62 (3) 324 (1989)

A. Georges and G. Kotliar Phys. Rev. B 45 (12) 6479 (1992)

Antoine Georges, Gabriel Kotliar, Werner Krauth, and Marcelo J. Rozenberg Rev. Mod. Phys. 68, 13 (1996)

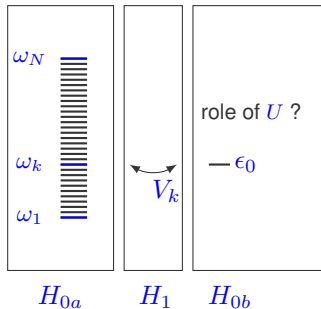
Anderson Hamiltonian in the DMFT



$$H_{\text{Anderson}} = \underbrace{\sum_{\sigma, k=1}^N \omega_k a_{k, \sigma}^+ a_{k \sigma}}_{H_{0a}} +$$

$$\underbrace{\epsilon_0(n_{\uparrow} + n_{\downarrow}) + U n_{\uparrow} n_{\downarrow}}_{H_{0b}}$$

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$$H_{\text{Anderson}} = \underbrace{\sum_{\sigma, k=1}^N \omega_k a_{k, \sigma}^+ a_{k \sigma}}_{H_{0a}} + \underbrace{\sum_{k, \sigma} V_k (a_{k, \sigma}^+ c_{\sigma} + c_{\sigma}^+ a_{k, \sigma})}_{H_1} + \underbrace{\epsilon_0 (n_{\uparrow} + n_{\downarrow}) + U n_{\uparrow} n_{\downarrow}}_{H_{0b}}$$

Anderson Hamiltonian with $V_k = 0$: Hubbard bands

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If $V_k = 0$,

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H_{0a} and H_{0b} are not coupled. Solution is equivalent to Hubbard model with $t = 0$: Hubbard bands.

Green's function: Non-Interacting Case

Green's Function

For a single orbital with energy ϵ_0 :

$$G_0(\omega) = \frac{1}{\omega - \epsilon_0 + i\eta}$$

Spectral Function

$$A(\omega) = -\frac{1}{\pi} \text{Im} G_0(\omega) = \delta(\omega - \epsilon_0)$$

Interacting Case

Full Green's Function

Includes self-energy $\Sigma(\omega)$:

$$G(\omega) = \frac{1}{\omega - \epsilon_0 - \Sigma(\omega) + i\eta}$$

Interpretation

$\Sigma(\omega)$ encodes interaction effects (Hubbard bands, renormalization).

Interacting Case (2)

Full Green's Function

Includes self-energy $\Sigma(\omega)$:

$$G(\omega) = (\omega I - H - \Sigma(\omega))^{-1}$$

Spectral function

$$A(\omega) = -\frac{1}{\pi} \text{Im} G(\omega)$$

One can show that the Green function is the correct quantity to describe photoemission spectra.

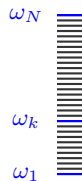
Anderson hamiltonian $U = 0$

The hamiltonian writes:

$$H_{\text{Anderson}} = \sum_{k,\sigma} \omega_k a_{k,\sigma}^+ a_{k\sigma} + \sum_{k,\sigma} V_k (a_{k,\sigma}^+ c_\sigma + c_\sigma^+ a_{k,\sigma}) + \epsilon_0 (n_\uparrow + n_\downarrow)$$

With only one bath state, H is:

$$\begin{pmatrix} \epsilon_0 & V_1 & \dots & V_k & \dots & V_N \\ V_1 & \omega_1 & 0 & 0 & 0 & 0 \\ \dots & 0 & \dots & 0 & 0 & 0 \\ V_k & 0 & 0 & \omega_k & 0 & 0 \\ \dots & 0 & 0 & 0 & \dots & 0 \\ V_N & 0 & 0 & 0 & 0 & \omega_N \end{pmatrix}$$



— ϵ_0



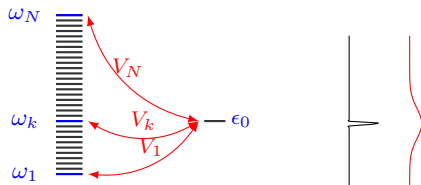
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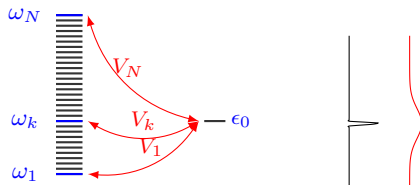
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This hamiltonian contains the hybridization of a single level ϵ_0 to other levels. The level with move and will be broadened by hybridization on other levels.

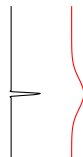
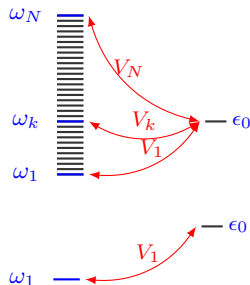
Anderson hamiltonian $U = 0$

H is:

$$\begin{pmatrix} \epsilon_0 & V_1 & \dots & V_k & \dots & V_N \\ V_1 & \omega_1 & 0 & 0 & 0 & 0 \\ \dots & 0 & \dots\dots\dots & 0 & 0 & 0 \\ V_k & 0 & 0 & \omega_k & 0 & 0 \\ \dots & 0 & 0 & 0 & \dots\dots\dots & 0 \\ V_N & 0 & 0 & 0 & 0 & \omega_N \end{pmatrix}$$

H is:

$$\begin{pmatrix} \epsilon_0 & V_1 \\ V_1 & \omega_1 \end{pmatrix}$$



Anderson hamiltonian $U = 0$

$$\begin{vmatrix} \epsilon_0 - \lambda & V_1 \\ V_1 & \omega_1 - \lambda \end{vmatrix} = 0$$

Anderson hamiltonian $U = 0$

$$\begin{vmatrix} \epsilon_0 - \lambda & V_1 \\ V_1 & \omega_1 - \lambda \end{vmatrix} = 0$$

Eigenvalues are the λ :

$$(\epsilon_0 - \lambda)(\omega_1 - \lambda) - V_1^2 = 0$$

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Let's now compute Green's function poles, using the following relation, which comes from the equation of motion of the Green's function (which comes from the time Schrodinger eq)

$$G = (\omega I - H)^{-1}$$

Anderson hamiltonian $U = 0$

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$$G = (\omega I - H)^{-1}$$

$$G = \begin{pmatrix} \omega - \epsilon_0 & V_1 \\ V_1 & \omega - \omega_1 \end{pmatrix}^{-1} = \frac{1}{(\omega - \epsilon_0)(\omega - \omega_1) - V_1^2} \begin{pmatrix} \omega - \epsilon_1 & -V_1 \\ -V_1 & \omega - \omega_0 \end{pmatrix}$$

The pole of G are indeed the eigenvalues of this two orbital model.

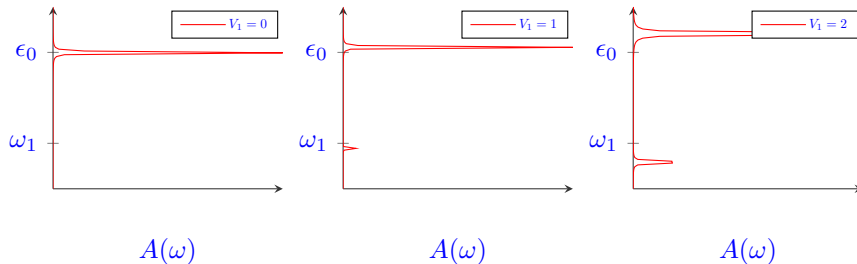
Let's now focus on the element of the Greens function on the correlated orbitals G_{00}

$$G_{00} = \frac{\omega - \epsilon_1}{(\omega - \epsilon_0)(\omega - \omega_1) - V_1^2} = \frac{1}{\omega - \epsilon_0 - \frac{V_1^2}{\omega - \omega_1}}$$

Spectral function

We can now compute the spectral function of this system by computing:

$$A(\omega) = -\frac{1}{\pi} \text{Im} G^R(\omega + i\delta) = -\frac{1}{\pi} \text{Im} \frac{1}{\omega + i\delta - \epsilon_0 - \frac{V_1^2}{\omega + i\delta - \omega_1}}$$



Anderson Hamiltonian with $U = 0$.

H is:

$$\begin{pmatrix} \epsilon_0 & V_1 & \dots & V_k & \dots & V_N \\ V_1 & \omega_1 & 0 & 0 & 0 & 0 \\ \dots & 0 & \dots & 0 & 0 & 0 \\ V_k & 0 & 0 & \omega_k & 0 & 0 \\ \dots & 0 & 0 & 0 & \dots & 0 \\ V_N & 0 & 0 & 0 & 0 & \omega_N \end{pmatrix}$$

$$G = (\omega I - H)^{-1}$$

We can inverse this matrix and compute the Green's function of the correlated orbital (Using $A^{-1} = \text{Com}(A)^T / \det A$ to inverse $I - H$). We obtain as a generalisation of the previous result:

$$G(\omega) = \frac{1}{\omega - \epsilon_0 - \Delta(\omega)} \quad \text{with} \quad \Delta(\omega) = \sum_k \frac{V_k^2}{\omega - \omega_k}$$

Where $\Delta(\omega)$ is called the hybridization function.

Anderson Hamiltonian with $U = 0$.

We can now compute the spectral function of this system by computing:

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We need

$$\Delta(\omega + i\delta) = \sum_k \frac{V_k^2}{\omega + i\delta - \omega_k}$$

$$\text{Re}\Delta = \sum_k \frac{V_k^2}{(\omega - \omega_k)^2 + \delta^2} \quad \text{Im}\Delta = \sum_k \frac{-\delta V_k^2}{(\omega - \omega_k)^2 + \delta^2}.$$

And with $\lim_{\delta \rightarrow 0} \frac{1}{\pi} \frac{\delta}{x^2 + \delta^2} = \delta(x)$

$$\text{Im}\Delta = -\pi \sum_k V_k^2 \delta(\omega - \omega_k) \simeq -\pi |V|^2 \rho(\omega)$$

Anderson Hamiltonian with $U = 0$.

We can now compute the spectral function of this system by computing:

$$A(\omega) = -\frac{1}{\pi} \text{Im} G^R(\omega + i\delta) = -\frac{1}{\pi} \text{Im} \frac{1}{\omega + i\delta - \epsilon_0 - \Delta(\omega + i\delta)}$$

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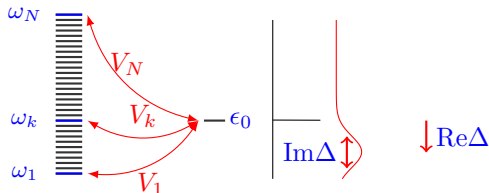
$$\Delta(\omega + i\delta) = \sum_k \frac{V_k^2}{\omega + i\delta - \omega_k}$$

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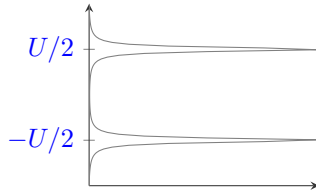
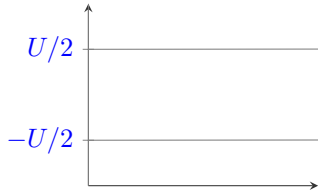
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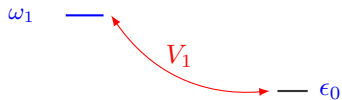
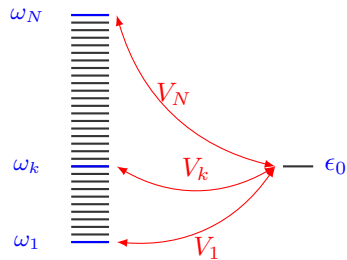
$$A(\omega) = -\frac{1}{\pi} \text{Im} G^R(\omega + i\delta) = -\frac{1}{\pi} \frac{\text{Im}\Delta}{(\omega - \epsilon_0 - \text{Re}\Delta)^2 + \text{Im}\Delta^2}$$



Anderson Hamiltonian; Hubbard band with hybridization ?



title



- Size of the Hilbert space ?
- If $N=2$ electrons ?
- If $U = \infty$?

Anderson Hamiltonian: one orbital for the bath.



Anderson Hamiltonian: one orbital for the bath.

ω_1 —

— ϵ_0

Anderson Hamiltonian: one orbital for the bath.

$$V = 0$$

$$\omega_1 \quad \uparrow$$

$$\uparrow \quad \epsilon_0$$

$$\omega_1 + \epsilon_0 \quad \text{—}$$

Anderson Hamiltonian: one orbital for the bath.

$$V = 0$$

$$\omega_1 \quad \text{---} \uparrow \downarrow$$

$$\text{---} \uparrow \quad \epsilon_0$$

$$\omega_1 + \epsilon_0 \quad \text{=====}$$

Anderson Hamiltonian: one orbital for the bath.

$$V = 0$$

$$\omega_1 \quad \uparrow\downarrow$$

$$\uparrow\downarrow \quad \epsilon_0$$

$$\omega_1 + \epsilon_0 \quad \equiv \equiv \equiv$$

Anderson Hamiltonian: one orbital for the bath.

$$V = 0$$

$$\omega_1 \uparrow$$

$$\downarrow \epsilon_0 \quad (S=0, S=1) \quad \omega_1 + \epsilon_0 \quad \equiv$$

Anderson Hamiltonian: one orbital for the bath.

$$V = 0$$

$$\omega_1 \quad \uparrow\downarrow$$

$$(S = 0)$$

$$2\omega_1$$

$$\text{—}$$

$$\text{— } \epsilon_0$$

$$(S = 0, S = 1) \quad \omega_1 + \epsilon_0$$

$$\text{====}$$

Anderson Hamiltonian: one orbital for the bath.

$$V = 0$$

$$\omega_1 \text{ —}$$

$$(S = 0)$$

$$2\omega_1 \text{ —}$$

$$\uparrow\downarrow \epsilon_0$$

$$(S = 0, S = 1) \omega_1 + \epsilon_0 \text{ } \text{=====}$$

$$2\epsilon_0 \text{ —}$$

Anderson Hamiltonian: one orbital for the bath.

$$2\epsilon_0 + U = \infty$$

$$V = 0$$

$$\omega_1 \text{ —}$$

$$(S = 0)$$

$$2\omega_1 \text{ —}$$

$$\uparrow\downarrow \epsilon_0$$

$$(S = 0, S = 1) \omega_1 + \epsilon_0 \text{ ———}$$

$$2\epsilon_0 \text{ —}$$

Anderson Hamiltonian: one orbital for the bath.

$$2\epsilon_0 + U = \infty$$

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ω_1 —

$(S = 0)$

$2\omega_1$ —

— ϵ_0

$(S = 0, S = 1)$ $\omega_1 + \epsilon_0$ ═══════

Anderson Hamiltonian: one orbital for the bath.

$$2\epsilon_0 + U = \infty$$

$$V = 0$$

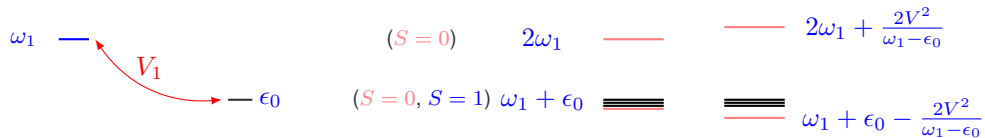


Anderson Hamiltonian: one orbital for the bath.

$$2\epsilon_0 + U = \infty$$

$$V = 0$$

$$V \neq 0$$

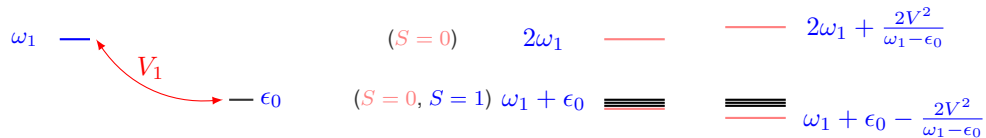


Anderson Hamiltonian: one orbital for the bath.

$$2\epsilon_0 + U = \infty$$

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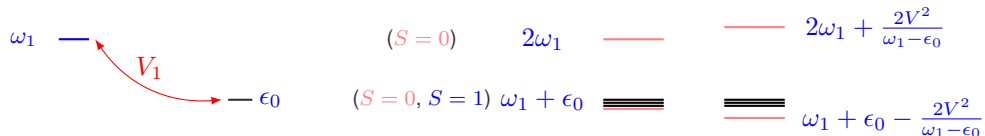
Because of hybridization, the ground state is a many body problem

Anderson Hamiltonian: one orbital for the bath.

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Because of hybridization, the ground state is a many body problem

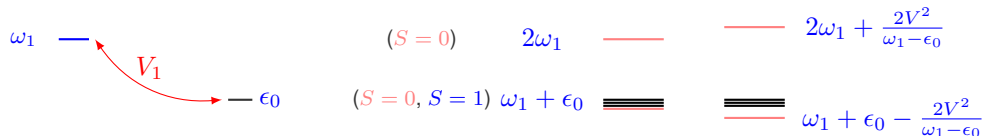
Because of U , the ground state is a many body problem

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Because of hybridization, the ground state is a many body problem

Because of U , the ground state is a many body problem

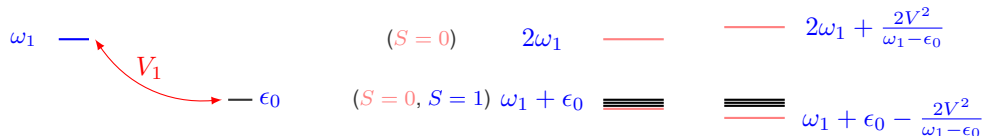
Because of U and hybridization, the ground state is not magnetic.

Anderson Hamiltonian: one orbital for the bath.

$$2\epsilon_0 + U = \infty$$

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$$V \neq 0$$



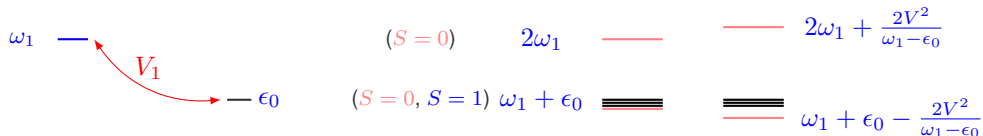
Because of hybridization, the ground state is a many body problem
 Because of U , the ground state is a many body problem
 Because of U and hybridization, the ground state is not magnetic.
 The formation of a singlet is the essence of the Kondo effect.

Anderson Hamiltonian: one orbital for the bath.

$$2\epsilon_0 + U = \infty$$

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$$V \neq 0$$



Because of hybridization, the ground state is a many body problem

Because of U , the ground state is a many body problem

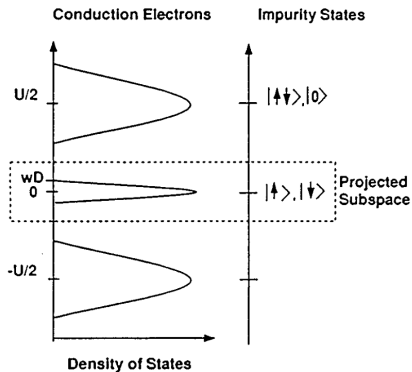
Because of U and hybridization, the ground state is not magnetic.

The formation of a singlet is the essence of the Kondo effect.

There is a low energy excitation $\simeq \frac{V^2}{\omega_1 - \epsilon_0}$

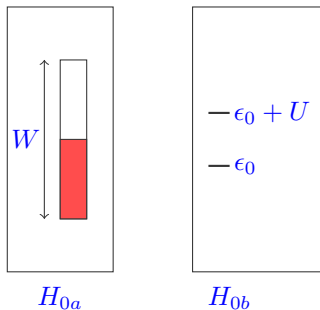
Fulde 1988

The Anderson model: 3 peak structure



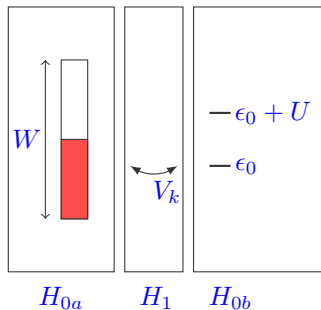
Hubbard bands are due to charge fluctuations (as in the atomic case)
Quasiparticle peak is linked to spin fluctuations (see Anderson molecule)
(from Georges et al RMP 1996)

The Anderson Hamiltonian (solved by CTQMC)



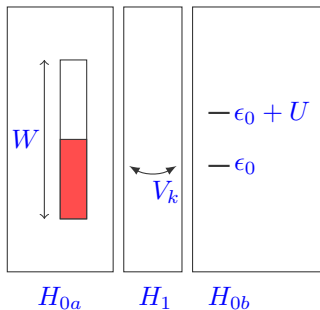
E Gull, AJ Millis, AI Lichtenstein, AN Rubtsov, M Troyer, P Werner Reviews of Modern Physics 83 (2), 349 (2011)

The Anderson Hamiltonian (solved by CTQMC)



E Gull, AJ Millis, AI Lichtenstein, AN Rubtsov, M Troyer, P Werner Reviews of Modern Physics 83 (2), 349 (2011)

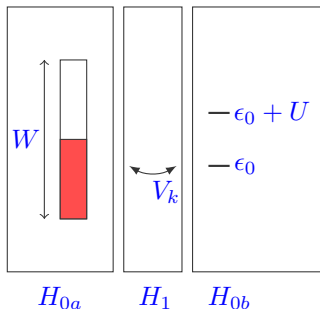
The Anderson Hamiltonian (solved by CTQMC)



$$H_{\text{Anderson}} = \underbrace{\sum \omega_k a_{k,\sigma}^+ a_{k\sigma}}_{H_{0a}} + \underbrace{\sum_{k,\sigma} V_k f a_{k,\sigma}^+ f_\sigma}_{H_1} + \underbrace{\sum_{\sigma} \epsilon_f f_\sigma^+ f_\sigma + U n_{f\uparrow} n_{f\downarrow}}_{H_{0b}}$$

E Gull, AJ Millis, AI Lichtenstein, AN Rubtsov, M Troyer, P Werner Reviews of Modern Physics 83 (2), 349 (2011)

The Anderson Hamiltonian (solved by CTQMC)



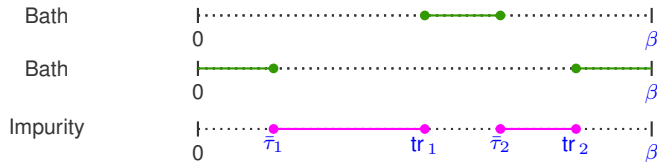
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Continuous Time Quantum Monte Carlo: Expansion as a function of H_1

[P. Werner, A. Comanac, L. de medici, M. Troyer and A. J. Millis Phys. Rev. Lett. 97, 076405 (2006)] E Gull, AJ Millis, Al Lichtenstein, AN Rubtsov, M Troyer, P Werner Reviews of Modern Physics 83 (2), 349 (2011)

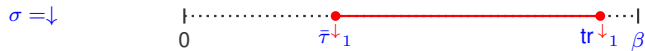
Hybridization expansion for Z_n

$$Z_n = \int_0^\beta d\tau_1 \dots \int_0^\beta d\tau_n \int_0^\beta d\bar{\tau}_1 \dots \int_0^\beta d\bar{\tau}_n \text{Tr} \left[\mathcal{T} \varepsilon^{-\beta H_0} \right. \\ \left. \sum_{k_1 \dots k_n, \bar{k}_1 \dots \bar{k}_n} \sum_{\sigma_1 \dots \sigma_n, \bar{\sigma}_1 \dots \bar{\sigma}_n} V_{k_1}^{\sigma_1} V_{\bar{k}_1}^{\bar{\sigma}_1 *} \dots V_{k_n}^{\sigma_n} V_{\bar{k}_n}^{\bar{\sigma}_n *} c_{k_n}^\dagger(\tau_n) d_{\sigma_n}(\tau_n) d_{\bar{\sigma}_n}^\dagger(\bar{\tau}_n) c_{\bar{k}_n}(\bar{\tau}_n) \dots c_{k_1}^\dagger(\tau_1) d_{\sigma_1}(\tau_1) d_{\bar{\sigma}_1}^\dagger(\bar{\tau}_1) c_{\bar{k}_1}(\bar{\tau}_1) \right]$$



Calculation of trace: need spins

$$\langle 00 | \varepsilon^{-(\beta H_d)} d_{\downarrow}(\text{tr}_{\uparrow}^{\downarrow}) d_{\uparrow}(\text{tr}_{\uparrow}^{\uparrow}) d_{\downarrow}^{\dagger}(\bar{\tau}_{\uparrow}^{\downarrow}) d_{\uparrow}^{\dagger}(\bar{\tau}_{\uparrow}^{\uparrow}) | 00 \rangle$$



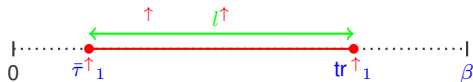
$$H_d = \sum_{\sigma} \varepsilon_0 d_{\sigma}^{\dagger} d_{\sigma} + U n_d^{\uparrow} n_d^{\downarrow}$$

$$\langle 00 | \varepsilon^{-(\beta H_d)} d_{\downarrow}(\text{tr}_{\uparrow}^{\downarrow}) d_{\uparrow}(\text{tr}_{\uparrow}^{\uparrow}) d_{\downarrow}^{\dagger}(\bar{\tau}_{\uparrow}^{\downarrow}) d_{\uparrow}^{\dagger}(\bar{\tau}_{\uparrow}^{\uparrow}) | 00 \rangle =$$

Calculation of trace: need spins

$$\langle 00 | \varepsilon^{-(\beta H_d)} d_{\downarrow}(\text{tr}_{\uparrow}^{\downarrow}) d_{\uparrow}(\text{tr}_{\uparrow}^{\uparrow}) d_{\downarrow}^{\dagger}(\bar{\tau}_{\uparrow}^{\downarrow}) d_{\uparrow}^{\dagger}(\bar{\tau}_{\uparrow}^{\uparrow}) | 00 \rangle$$

$\sigma = \uparrow$



$\sigma = \downarrow$

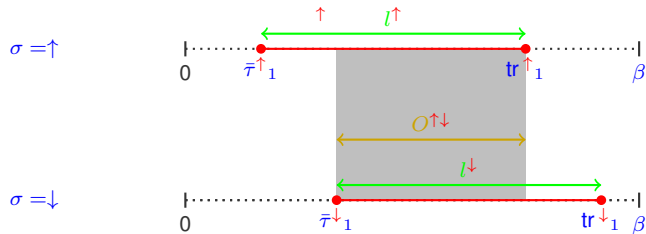


$$H_d = \sum_{\sigma} \varepsilon_0 d_{\sigma}^{\dagger} d_{\sigma} + U n_d^{\uparrow} n_d^{\downarrow}$$

$$\langle 00 | \varepsilon^{-(\beta H_d)} d_{\downarrow}(\text{tr}_{\uparrow}^{\downarrow}) d_{\uparrow}(\text{tr}_{\uparrow}^{\uparrow}) d_{\downarrow}^{\dagger}(\bar{\tau}_{\uparrow}^{\downarrow}) d_{\uparrow}^{\dagger}(\bar{\tau}_{\uparrow}^{\uparrow}) | 00 \rangle = \boxed{\exp \left[-\varepsilon_0 (l^{\uparrow} + l^{\downarrow}) \right]}$$

Calculation of trace: need spins

$$\langle 00 | \varepsilon^{-(\beta H_d)} d_{\downarrow}(\text{tr}_{\uparrow}^{\downarrow}) d_{\uparrow}(\text{tr}_{\uparrow}^{\uparrow}) d_{\downarrow}^{\dagger}(\bar{\tau}_{\uparrow}^{\downarrow}) d_{\uparrow}^{\dagger}(\bar{\tau}_{\uparrow}^{\uparrow}) | 00 \rangle$$

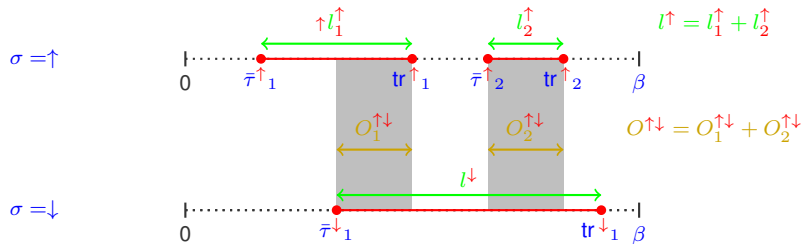


$$H_d = \sum_{\sigma} \varepsilon_0 d_{\sigma}^{\dagger} d_{\sigma} + U n_d^{\uparrow} n_d^{\downarrow}$$

$$\langle 00 | \varepsilon^{-(\beta H_d)} d_{\downarrow}(\text{tr}_{\uparrow}^{\downarrow}) d_{\uparrow}(\text{tr}_{\uparrow}^{\uparrow}) d_{\downarrow}^{\dagger}(\bar{\tau}_{\uparrow}^{\downarrow}) d_{\uparrow}^{\dagger}(\bar{\tau}_{\uparrow}^{\uparrow}) | 00 \rangle = \exp \left[-\varepsilon_0 (l^{\uparrow} + l^{\downarrow}) - U O^{\uparrow\downarrow} \right]$$

Calculation of trace: need spins

$$\langle 00 | \varepsilon^{-(\beta H_d)} d_{\downarrow}(\text{tr}_{\uparrow 1}^{\downarrow}) d_{\uparrow}(\text{tr}_{\uparrow 2}^{\uparrow}) d_{\uparrow}^{\dagger}(\bar{\tau}_{\uparrow 2}^{\uparrow}) d_{\uparrow}(\text{tr}_{\uparrow 1}^{\uparrow}) d_{\downarrow}^{\dagger}(\bar{\tau}_{\uparrow 1}^{\downarrow}) d_{\uparrow}^{\dagger}(\bar{\tau}_{\uparrow 1}^{\uparrow}) | 00 \rangle$$

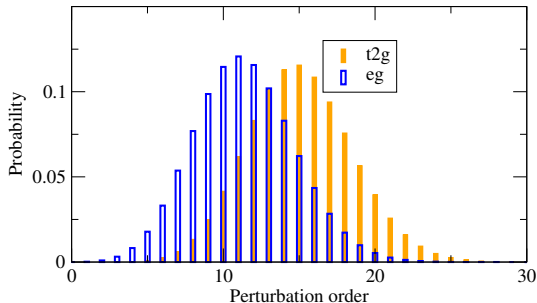
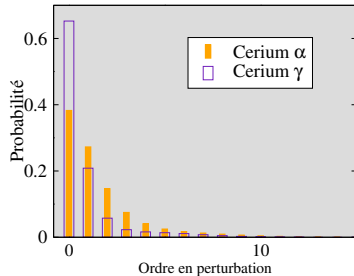


$$H_d = \sum_{\sigma} \varepsilon_0 d_{\sigma}^{\dagger} d_{\sigma} + U n_d^{\uparrow} n_d^{\downarrow}$$

$$\langle 00 | \varepsilon^{-(\beta H_d)} d_{\downarrow}(\text{tr}_{\uparrow 1}^{\downarrow}) d_{\uparrow}(\text{tr}_{\uparrow 2}^{\uparrow}) d_{\uparrow}^{\dagger}(\bar{\tau}_{\uparrow 2}^{\uparrow}) d_{\uparrow}(\text{tr}_{\uparrow 1}^{\uparrow}) d_{\downarrow}^{\dagger}(\bar{\tau}_{\uparrow 1}^{\downarrow}) d_{\uparrow}^{\dagger}(\bar{\tau}_{\uparrow 1}^{\uparrow}) | 00 \rangle = \exp \left[-\varepsilon_0 (l^{\uparrow} + l^{\downarrow}) - U O^{\uparrow\downarrow} \right]$$

Comparison between Iron and Cerium

- d orbitals in iron are much diffuse than f orbitals in cerium.
- V_k is thus much larger
- The expansion as a function in V_k needs more term in iron in comparison to cerium.



The self consistency condition and the DMFT loop

The equation of Motion of the Green's function is

$$(\omega - H - \Sigma)G = 1$$

It is an equation for operators and it directly comes from the time dependent Schrödinger Eq. for creation and annihilation operators. One can insert the closure relation for Bloch states which is

$$\sum_{k'} |\psi_{k'}\rangle \langle \psi_{k'}| = 1$$

We thus have:

$$(\omega - H - \Sigma) \sum_{k'} |\psi_{k'}\rangle \langle \psi_{k'}| G = 1$$

We can project on the left and right by ψ_k :

$$\langle \psi_k | (\omega - H - \Sigma) \sum_{k'} |\psi_{k'}\rangle \langle \psi_{k'}| G | \psi_k \rangle = 1$$

thus

$$[\omega - \epsilon_{\mathbf{k}} - \Sigma_{\mathbf{k}}(\omega)] G_{\mathbf{k}}(\omega)$$

the lattice Green's function for the Hubbard model is written:

$$G_{\mathbf{k}}(\omega) = \frac{1}{\omega - \epsilon_{\mathbf{k}} - \Sigma_{\mathbf{k}}(\omega)}$$

Green's function of the lattice

The Green's function of the lattice in real space writes:

$$G_{ij}^{\text{local}}(\omega) = \frac{1}{N} \sum_{\mathbf{k}} e^{ik(T_i - T_j)} G_{\mathbf{k}}(\omega) = \frac{1}{N} \sum_{\mathbf{k}} G_{\mathbf{k}}(\omega)$$

The local Green's function of the lattice is

$$G_{ii}^{\text{local}}(\omega) = \frac{1}{N} \sum_{\mathbf{k}} e^{ik(T_i - T_i)} G_{\mathbf{k}}(\omega) = \frac{1}{N} \sum_{\mathbf{k}} G_{\mathbf{k}}(\omega)$$

$$G_{ii}^{\text{local}}(\omega) = \frac{1}{N} \sum_{\mathbf{k}} \frac{1}{\omega - \epsilon_{\mathbf{k}} - \Sigma_{\mathbf{k}}(\omega)}$$

Green's function of the Anderson model

- From the equation of motion:

$$G = (\omega I - H - \Sigma)^{-1}$$

with $H + \Sigma$ is:

$$\begin{pmatrix} \epsilon_0 + \Sigma & V_1 & \dots & V_k & \dots & V_N \\ V_1 & \omega_1 & 0 & 0 & 0 & 0 \\ \dots & 0 & \dots & 0 & 0 & 0 \\ V_k & 0 & 0 & \omega_k & 0 & 0 \\ \dots & 0 & 0 & 0 & \dots & 0 \\ V_N & 0 & 0 & 0 & 0 & \omega_N \end{pmatrix}$$

we can show using exactly the same derivation as for $U = 0$:

$$G_{\text{Anderson}}(\omega) = \frac{1}{\omega - \epsilon_0 - \Delta(\omega) - \Sigma(\omega)}$$

Green's function of the Anderson model

- From the equation of motion:

$$G = (\omega I - H - \Sigma)^{-1}$$

with $H + \Sigma$ is:

$$\begin{pmatrix} \epsilon_0 + \Sigma & V_1 & \dots & V_k & \dots & V_N \\ V_1 & \omega_1 & 0 & 0 & 0 & 0 \\ \dots & 0 & \dots & 0 & 0 & 0 \\ V_k & 0 & 0 & \omega_k & 0 & 0 \\ \dots & 0 & 0 & 0 & \dots & 0 \\ V_N & 0 & 0 & 0 & 0 & \omega_N \end{pmatrix}$$

we can show using exactly the same derivation as for $U = 0$:

$$G_{\text{Anderson}}(\omega) = \frac{1}{\omega - \epsilon_0 - \Delta(\omega) - \Sigma(\omega)}$$

- or equivalently using the Dyson equation using the expression of $\mathcal{G}_0^{-1}$

$$G^{-1} = \mathcal{G}_0^{-1} - \Sigma = \omega I - H - \Delta - \Sigma$$

$$G_{\text{Anderson}}(\omega) = \frac{1}{\omega - \epsilon_0 - \Delta(\omega) - \Sigma(\omega)}$$

The self consistency condition and the DMFT loop

For one atom in the solid, the local Green's function is

$$G_{ii}^{\text{local}}(\omega) = \frac{1}{N} \sum_{\mathbf{k}} \frac{1}{\omega - \epsilon_{\mathbf{k}} - \Sigma_{\mathbf{k}}(\omega)}$$

For the Anderson model, the correlated orbital Green's function is:

$$G_{\text{Anderson}}(\omega) = \frac{1}{\omega - \epsilon_0 - \Delta(\omega) - \Sigma(\omega)}$$

What could be the self-consistency relation ?

The self consistency condition and the DMFT loop

For one atom in the solid, the local Green's function is

$$G_{ii}^{\text{local}}(\omega) = \frac{1}{N} \sum_{\mathbf{k}} \frac{1}{\omega - \epsilon_{\mathbf{k}} - \Sigma_{\mathbf{k}}(\omega)}$$

For the Anderson model, the correlated orbital Green's function is:

$$G_{\text{Anderson}}(\omega) = \frac{1}{\omega - \epsilon_0 - \Delta(\omega) - \Sigma(\omega)}$$

Identity of spectral functions for the lattice and for the Anderson model

$$-\frac{1}{\pi} \text{Im}[G_{ii}^{\text{local}}(\omega)] = -\frac{1}{\pi} \text{Im}[G_{\text{Anderson}}]$$

The self consistency condition and the DMFT loop

For one atom in the solid, the local Green's function is

$$G_{ii}^{\text{local}}(\omega) = \frac{1}{N} \sum_{\mathbf{k}} \frac{1}{\omega - \epsilon_{\mathbf{k}} - \Sigma_{\mathbf{k}}(\omega)}$$

For the Anderson model, the correlated orbital Green's function is:

$$G_{\text{Anderson}}(\omega) = \frac{1}{\omega - \epsilon_0 - \Delta(\omega) - \Sigma(\omega)}$$

The DMFT idea is to identify the two Green's function and the Self-energies:

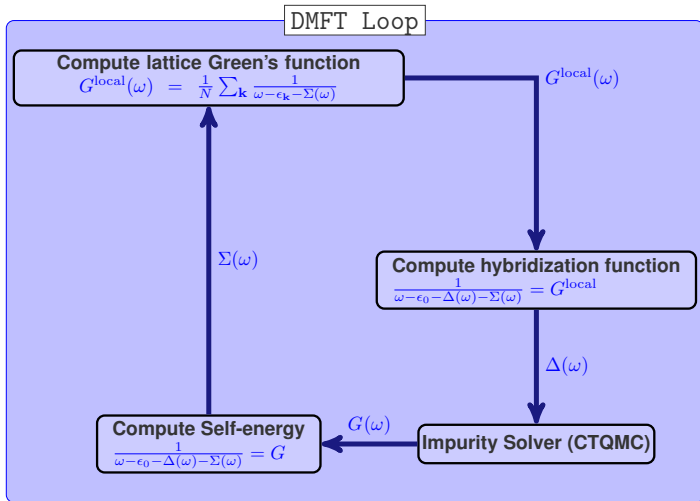
$$G_{ii}^{\text{local}}(\omega) = G_{\text{Anderson}}(\omega) \quad \Rightarrow \quad \frac{1}{N} \sum_{\mathbf{k}} \frac{1}{\omega - \epsilon_{\mathbf{k}} - \Sigma(\omega)} = \frac{1}{\omega - \epsilon_0 - \Delta(\omega) - \Sigma(\omega)}$$

DMFT scheme

$$G_{ii}^{\text{local}}(\omega) = \frac{1}{N} \sum_{\mathbf{k}} \frac{1}{\omega - \epsilon_{\mathbf{k}} - \Sigma_{\mathbf{k}}(\omega)}$$

$$G_{\text{Anderson}}(\omega) = \frac{1}{\omega - \epsilon_0 - \Delta(\omega) - \Sigma(\omega)}$$

$$G_{ii}^{\text{local}}(\omega) = G_{\text{Anderson}}(\omega)$$



The self consistency condition and the DMFT loop

$$\frac{1}{N} \sum_k \frac{1}{\omega - \epsilon_{\mathbf{k}} - \Sigma(\omega)} = \frac{1}{\omega - \epsilon_0 - \Delta(\omega) - \Sigma(\omega)}$$

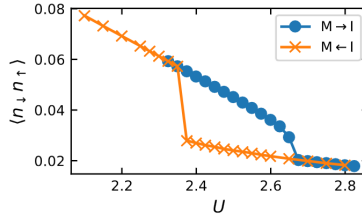
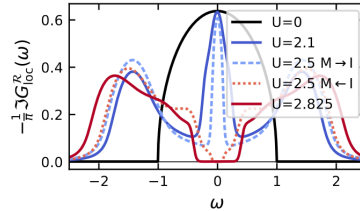
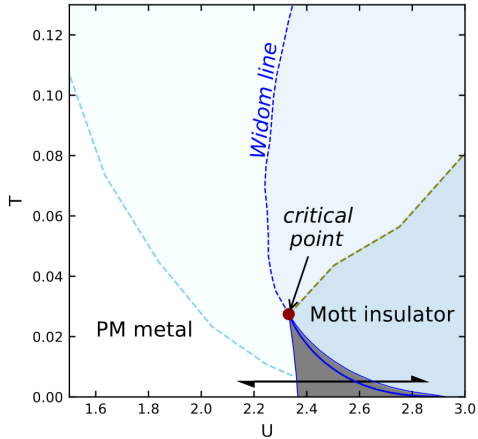
- If $t = 0$, then $\epsilon_{\mathbf{k}} = \epsilon_0$ and $\Delta = 0$ thus, the self-consistency is always fulfilled. The Anderson model can be solved only one time to give the exact solution.

$$\frac{1}{N} \sum_k \frac{1}{\omega - \epsilon_0 - \Sigma(\omega)} = \frac{1}{\omega - \epsilon_0 - \Sigma(\omega)}$$

- If $U = 0$, then the lattice Green's function and the local Green's function are exact.

$$\frac{1}{N} \sum_k \frac{1}{\omega - \epsilon_{\mathbf{k}}} = \frac{1}{\omega - \epsilon_0 - \Delta(\omega)}$$

Phase diagram of Hubbard model in DMFT



Watznböck et al Scipost (2022)

Phase diagram of Hubbard model in DMFT

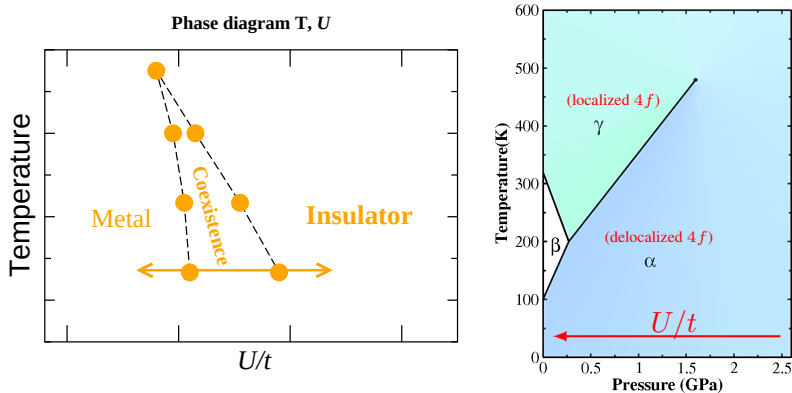


FIGURE: Phase diagram of the Hubbard model in DMFT compared to phase diagram of Cerium.

Phase diagram of Hubbard model in DMFT

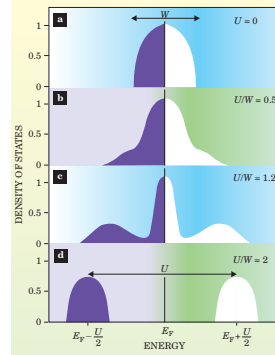
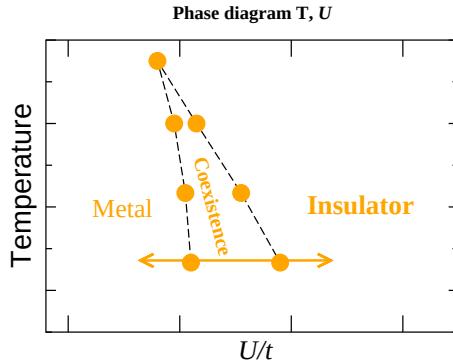
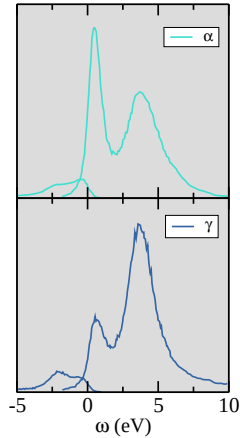
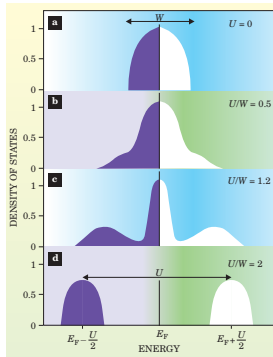


FIGURE: Phase diagram of the Hubbard model in DMFT and evolution of the spectral function

Spectral function of Hubbard model in DMFT



[G.Kotliar *et al* Phys. Today, AIP, 57, 53-59 (2004)]

[E. Weschke, *et al* Phys. Rev. B 44, 8304 (1991)
M. Gironi, *et al* Phys. Rev. B 55, 2056 (1997)]

DFT : The Kohn Sham Equations

- The total energy expression as a function of $\phi(\mathbf{r})$ is

$$E_{v_{\text{ext}}}[n(\mathbf{r})] = - \sum_i \int \phi_i^*(\mathbf{r}) \frac{\nabla^2}{2} \phi_i(\mathbf{r}) d\mathbf{r} + \int d\mathbf{r} v_{\text{ext}}(\mathbf{r}) n(\mathbf{r}) + E_{\text{Hartree}}[n(\mathbf{r})] + E_{\text{xc}}[n(\mathbf{r})]$$

and can be minimized.

- One obtains the one electron Kohn-Sham Equations

$$\left[-\frac{\nabla^2}{2} + V_{\text{KS}}(\mathbf{r}) \right] \phi_i(\mathbf{r}) = \epsilon_i \phi_i(\mathbf{r})$$

- The effective Kohn Sham potential $V_{\text{KS}}(\mathbf{r})$ is defined as a the functional derivative of the last three terms of the energy:

$$V_{\text{KS}}(\mathbf{r}) = v_{\text{ext}}(\mathbf{r}) + \int d\mathbf{r}' \frac{n(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} + v_{\text{xc}}[n(\mathbf{r})](\mathbf{r})$$

- The exchange and correlation potential $v_{\text{xc}}(\mathbf{r})$ is

$$v_{\text{xc}}(\mathbf{r}) = \frac{\delta E_{\text{xc}}[n(\mathbf{r})]}{\delta n(\mathbf{r})}$$

From DFT to DFT+DMFT

DMFT

DFT + DMFT

$$G_k^{\text{lattice}}(\omega) = \frac{1}{\omega - \epsilon_{\mathbf{k}} - \Sigma_{\mathbf{k}}(\omega)}$$

$$G_{ii}^{\text{local}}(\omega) = \frac{1}{N} \sum_{\mathbf{k}} G_k^{\text{lattice}}(\omega)$$

For the Anderson model, the correlated orbital Green's function is:

$$G_{\text{Anderson}}(\omega) = \frac{1}{\omega - \epsilon_0 - \Delta(\omega) - \Sigma(\omega)}$$

From DFT to DFT+DMFT

DMFT

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DFT + DMFT

- ϵ_k = Kohn Sham eigenvalues - double counting correction

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DMFT

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 $\Rightarrow G$ and Σ are matrices

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- Need a projection from Kohn Sham states to correlated orbitals. $\langle \Psi_{nk} | \chi_{lm} \rangle$

From DFT to DFT+DMFT

DMFT

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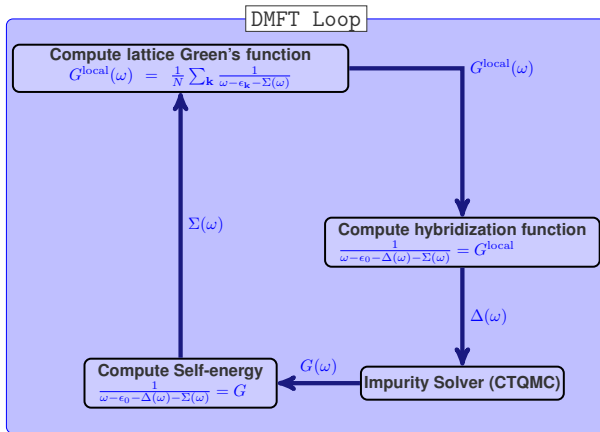
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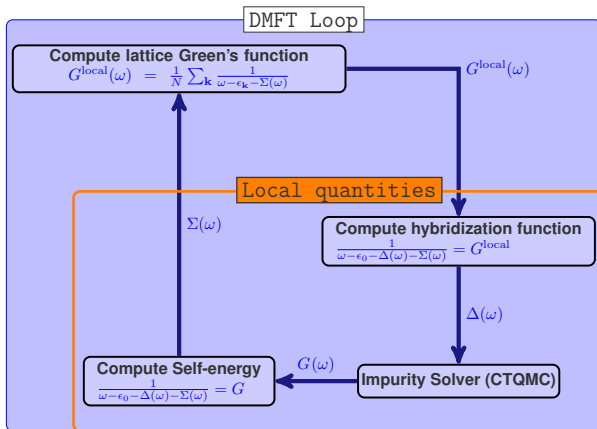
DFT + DMFT

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- The local Green's function is defined on correlated orbitals.
 $\Rightarrow G$ and Σ are matrices
- Need a projection from Kohn Sham states to correlated orbitals. $\langle \Psi_{nk} | \chi_{lm} \rangle$
- The Anderson model is multiorbital, e.g. 10 (d) or 14 (f) orbitals.

The DMFT Loop



The DMFT Loop



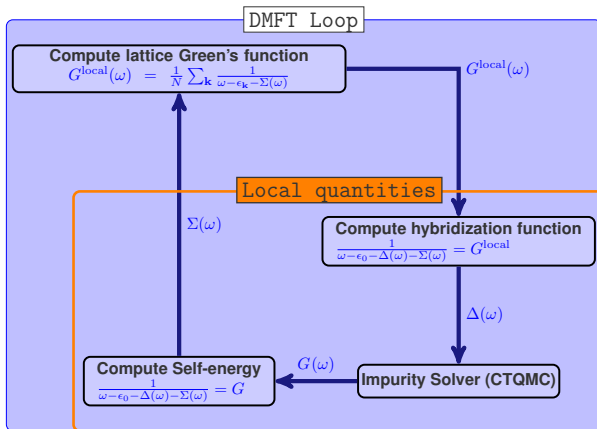
Reminder about Hubbard model

- Local quantities are expressed in a basis of correlated orbitals.

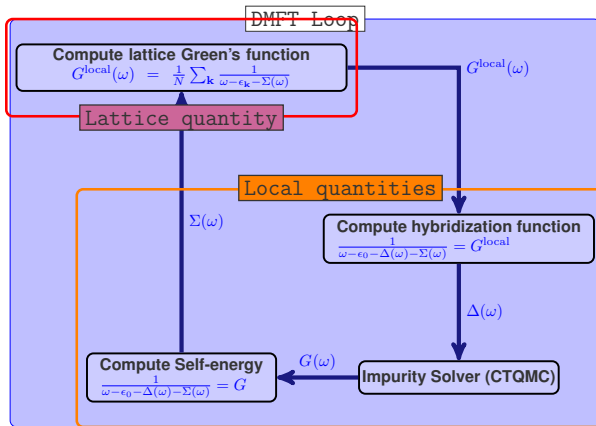
$$\hat{\Sigma}(\omega) = \sum_{\mathbf{T}} |\chi_{\mathbf{T}}\rangle \Sigma(\omega) \langle \chi_{\mathbf{T}}|$$

where $\chi_{\mathbf{T}}$ is the unique orbital on the atom at $\mathbf{T}$ and $\mathbf{T}$ are the lattice vectors.

The DMFT Loop



The DMFT Loop



Reminder about Hubbard model

- Lattice Green's functions are expressed in Bloch eigenvectors of the non interacting Hamiltonian ($\epsilon_{\mathbf{k}} = \frac{1}{N} \sum_{ij} t_{ij} e^{-i\mathbf{k}(\mathbf{T}_i - \mathbf{T}_j)}$)

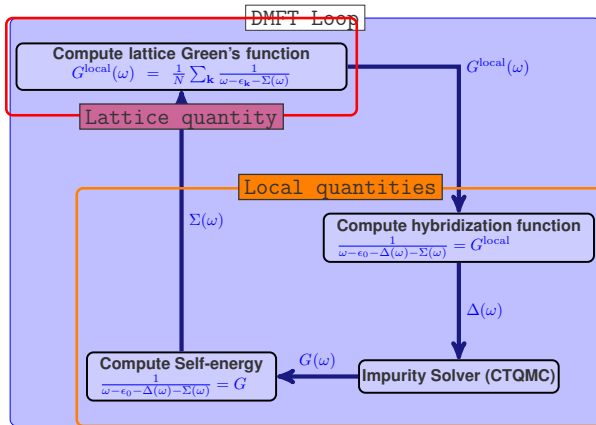
$$\hat{H}_0 = |\chi_{\mathbf{k}}\rangle \epsilon_{\mathbf{k}} \langle \chi_{\mathbf{k}}|$$

with

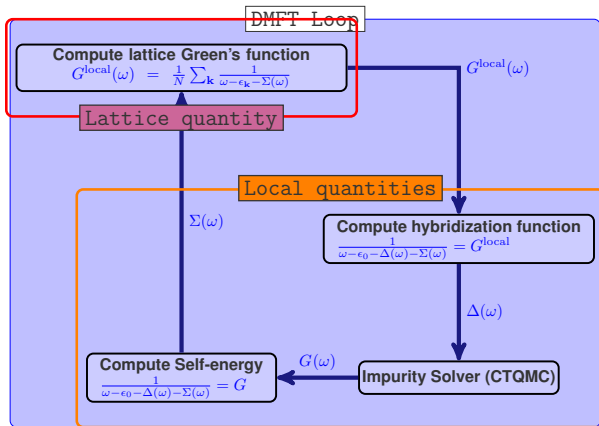
$$|\chi_{\mathbf{k}}\rangle = \frac{1}{\sqrt{N}} \sum_{\mathbf{T}} |\chi_{\mathbf{T}}\rangle e^{i\mathbf{k}\mathbf{T}}$$

where $\mathbf{T}$ are lattice vectors.

The DMFT Loop

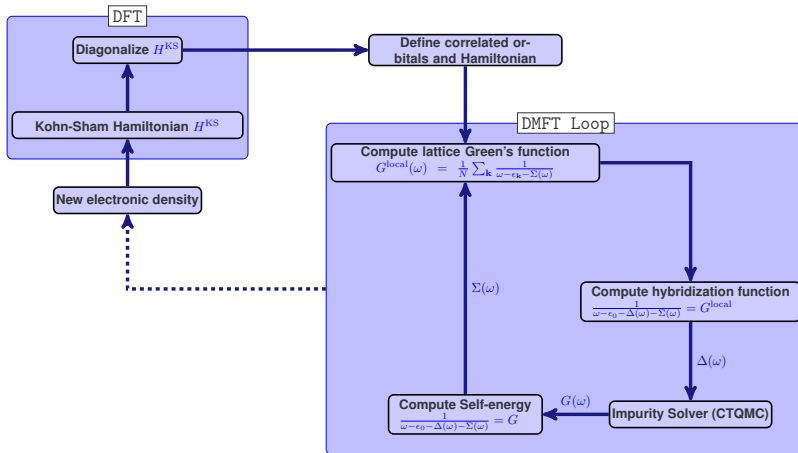


The DMFT Loop

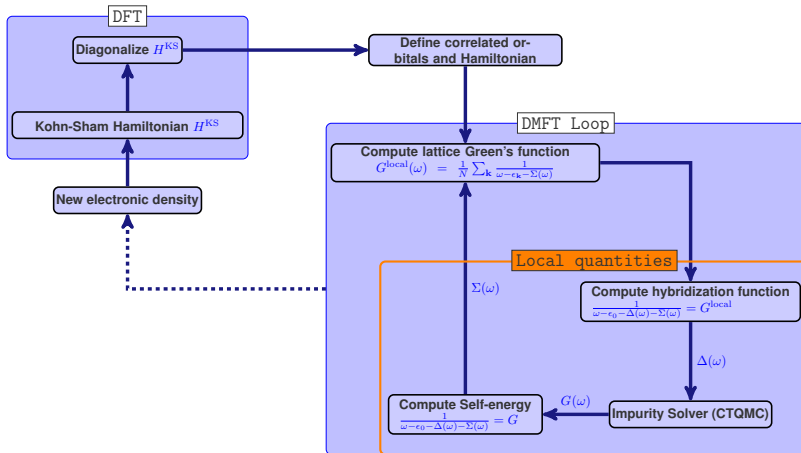


How to apply such idea for a real solid, with both **strongly interacting orbitals**, and **weakly interacting orbitals**

DFT+DMFT scheme



DFT+DMFT scheme



Correlated orbitals: what are they

For d orbitals ($l=2$), $m \in \{-2, -1, 0, 1, 2\}$, a correlated atomic orbital writes:

$$\chi_m(\mathbf{r}) = \langle \mathbf{r} | \chi_m \rangle = R(r)Y_m(\theta, \phi)$$

- The angular part $Y_m(\theta, \phi)$ is well defined.
- The radial part $R(r)$ is not defined in a solid.

In a first step, let's assume that we have found a good choice for $R(r)$ and thus $|\chi_m\rangle$ is defined.

In a real system in DFT+DMFT

- Local Quantity are expressed in a basis of correlated orbitals.

$$\hat{\Sigma}(\omega) = \sum_{m,m',\mathbf{T}} |\chi_{\mathbf{T}m}^{\mathbf{R}}\rangle \Sigma_{m,m'}(\omega) \langle \chi_{\mathbf{T}m'}^{\mathbf{R}}|$$

where $m, m' \in -l, \dots, l$ and $\chi_{\mathbf{T}m}^{\mathbf{R}}$ is an orbital whose angular part is Y_{lm} . Note that $\Sigma_{m,m'}$ can be a matrix and have non diagonal elements.

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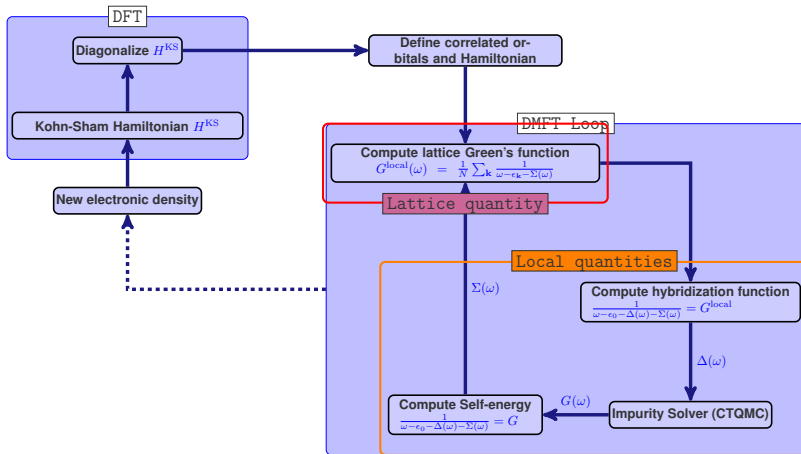
where $m, m' \in -l, \dots, l$ and $\chi_{\mathbf{T}m}^{\mathbf{R}}$ is an orbital whose angular part is Y_{lm} . Note that $\Sigma_{m,m'}$ can be a matrix and have non diagonal elements.

[for Hubbard model

$$\hat{\Sigma}(\omega) = \sum_{\mathbf{T}} |\chi_{\mathbf{T}}\rangle \Sigma(\omega) \langle \chi_{\mathbf{T}}|$$

]

DFT+DMFT scheme



In a real system in DFT+DMFT

- DFT Hamiltonian and thus lattice Green's functions are more easily expressed in Bloch eigenvectors of the DFT Kohn Sham Hamiltonian.

$$\hat{H}_{\text{KS}} = |\Psi_{\mathbf{k}\nu}\rangle \epsilon_{\mathbf{k}\nu} \langle \Psi_{\mathbf{k}\nu}|$$

Where $\Psi_{\mathbf{k}\nu}$ are one electron Kohn Sham wave function for the k-point $\mathbf{k}$ and band number ν .

$\Psi_{\mathbf{k}\nu}$ contains both correlated atomic orbitals and other non correlated orbitals.

(Reminder: in the Hubbard model, the analogue of $\Psi_{\mathbf{k}\nu}$ was just the Bloch transform of atomic orbitals.)

In a real system in DFT+DMFT

- Local quantities such as $\hat{\Sigma}$ can be computed in the Bloch basis:

$$\Sigma_{\nu\nu'\mathbf{k}}(\omega) = \langle \Psi_{\mathbf{k}\nu} | \hat{\Sigma}(\omega) | \Psi_{\mathbf{k}\nu'} \rangle = \sum_{m,m',\mathbf{T}} \langle \Psi_{\mathbf{k}\nu} | \chi_{\mathbf{T}m}^{\mathbf{R}} \rangle \Sigma_{mm'}(\omega) \langle \chi_{\mathbf{T}m'}^{\mathbf{R}} | \Psi_{\mathbf{k}\nu'} \rangle$$

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Using $|\chi_{\mathbf{T}m}^{\mathbf{R}}\rangle = \frac{1}{\sqrt{N}} \sum_{\mathbf{k}} |\chi_{\mathbf{k}m}^{\mathbf{R}}\rangle e^{-i\mathbf{k}\mathbf{T}}$, one arrive to

$$\Sigma_{\nu\nu'\mathbf{k}}(\omega) = \sum_{m,m'} \langle \Psi_{\mathbf{k}\nu} | \chi_{\mathbf{k}m}^{\mathbf{R}} \rangle \Sigma_{mm'}(\omega) \langle \chi_{\mathbf{k}m'}^{\mathbf{R}} | \Psi_{\mathbf{k}\nu'} \rangle$$

Note that now, the self-energy is a $\mathbf{k}$ dependent quantity, because of weight of orbitals over Kohn Sham states depends on the k-point.

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- From the Self energy, the full lattice Green's function in the Kohn Sham basis

$$G^{-1}(\omega) = G_{\text{KS}}^{-1}(\omega) - \Delta \Sigma(\omega) = (\omega - H_{\text{KS}}) \mathcal{I} - \Delta \Sigma(\omega)$$

$$G_{\nu\nu'\mathbf{k}}(\omega) = [\omega - \epsilon_{\mathbf{k}\nu} - \Delta \Sigma_{\nu\nu'\mathbf{k}}(\omega)]_{\nu\nu'\mathbf{k}}^{-1}$$

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In a real system in DFT+DMFT

- Using the operator expression of the lattice Green's function

$$\hat{G}(\omega) = \sum_{\nu\nu'\mathbf{k}} |\Psi_{\mathbf{k}\nu}\rangle G_{\nu\nu'\mathbf{k}}(\omega) \langle\Psi_{\mathbf{k}\nu'}|$$

In a real system in DFT+DMFT

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$$\hat{G}(\omega) = \sum_{\nu\nu'\mathbf{k}} |\Psi_{\mathbf{k}\nu}\rangle G_{\nu\nu'\mathbf{k}}(\omega) \langle\Psi_{\mathbf{k}\nu'}|$$

- One can write the local Green's function as:

$$G_{mm'}(\omega) = \sum_{\nu\nu'\mathbf{k}} \langle\chi_{\mathbf{T}m}^{\mathbf{R}}|\Psi_{\mathbf{k}\nu}\rangle G_{\nu\nu'\mathbf{k}}(\omega) \langle\Psi_{\mathbf{k}\nu'}|\chi_{\mathbf{T}m'}^{\mathbf{R}}\rangle$$

for Hubbard model $G(\omega) = \sum_{\mathbf{k}} G_{\mathbf{k}}(\omega)$

From DFT to DFT+DMFT

DMFT

$$G_k^{\text{lattice}}(\omega) = \frac{1}{\omega - \epsilon_{\mathbf{k}} - \Sigma_{\mathbf{k}}(\omega)}$$

$$G^{\text{local}}(\omega) = \frac{1}{N} \sum_{\mathbf{k}} G_k^{\text{lattice}}(\omega)$$

For the Anderson model, the correlated orbital Green's function is:

$$G_{\text{Anderson}}(\omega) = \frac{1}{\omega - \epsilon_0 - \Delta(\omega) - \Sigma(\omega)}$$

DFT + DMFT

$$G_{n,n'\mathbf{k}}^{\text{lattice}}(\omega) = \frac{1}{\omega - \epsilon_{n\mathbf{k}} - \Sigma_{nn'\mathbf{k}}(\omega)}$$

$$G_{mm'}^{\text{local}}(\omega) = \sum_{\mathbf{k}, n, n'} \langle \chi_{\mathbf{k}m} | \Psi_{\mathbf{k}n} \rangle G_{nn'}(\mathbf{k}, \omega) \langle \Psi_{\mathbf{k}n'} | \chi_{\mathbf{k}m'} \rangle$$

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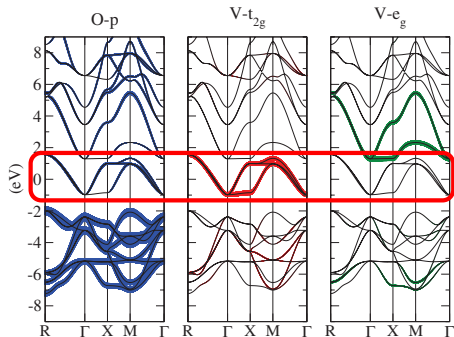
$$G_{mm'}^{\text{Anderson}}(\omega) = [\omega I - E_0 - \Delta(\omega) - \Sigma(\omega)]^{-1} \big|_{mm'}$$

Definition of correlated orbitals: example of SrVO_3 .

Need a projection from Kohn Sham states to correlated orbitals. $\langle \Psi_{nk} | \chi_{lm} \rangle$
A possible choice for correlation orbitals is Wannier functions.

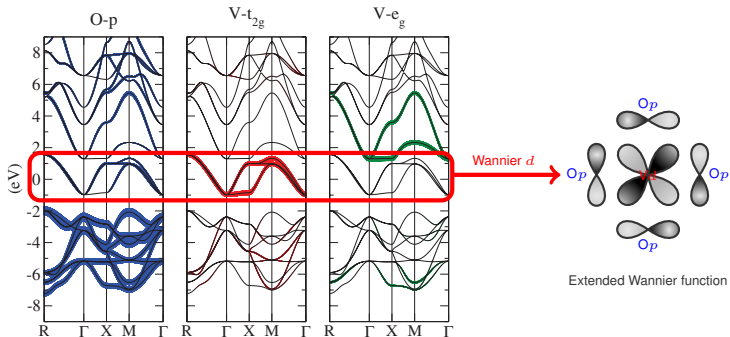
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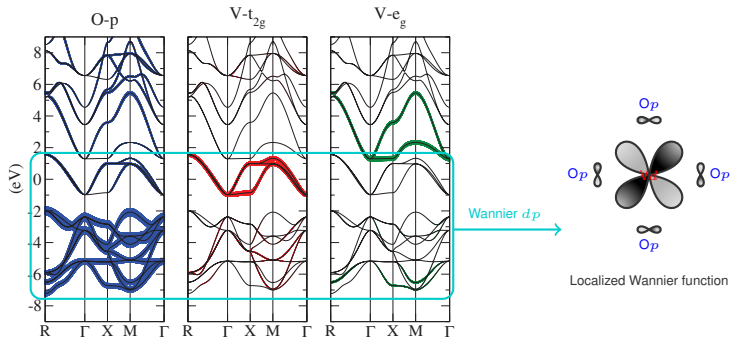
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Wannier orbitals are made from V dt_{2g} bands so, they are not pure dt_{2g} orbitals because of the hybridization

Definition of correlated orbitals: example of SrVO_3 .

Need a projection from Kohn Sham states to correlated orbitals. $\langle \Psi_{nk} | \chi_{lm} \rangle$
A possible choice for correlation orbitals is Wannier functions.



Wannier orbitals are made from V dt_{2g} bands and O p bands so, they are closer to dt_{2g} orbitals because more t_{2g} character is taken into account.

Double counting of interactions

Double counting corrections: Atomic limit (or Full localized limit) [Lichtenstein(1995), Anisimov (1991)]:

$$E_{\text{dc}}^{\text{FLL}} = \sum_t \left(\frac{U}{2} N(N-1) - \sum_{\sigma} \frac{J}{2} N^{\sigma}(N^{\sigma}-1) \right)$$

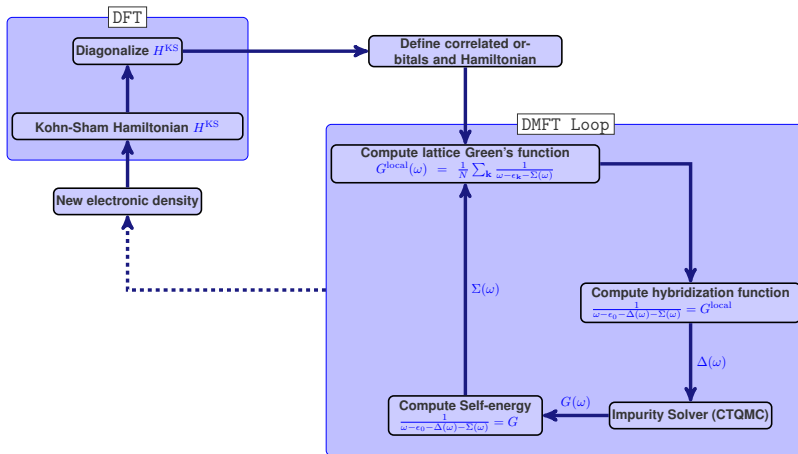
Around mean field version [Czyzyk(1994)] (delocalized limit):

$$E_{\text{dc}}^{\text{AMF}} = \sum_t \left(U N_{\uparrow} N_{\downarrow} + \frac{1}{2} (N_{\uparrow}^2 + N_{\downarrow}^2) \frac{2l}{2l+1} (U - J) \right)$$

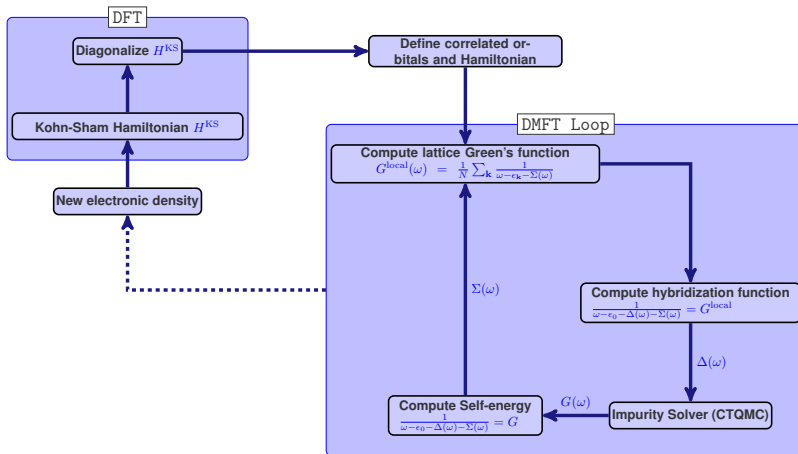
(Made to correct the delocalized limit.)

$$G_{\text{Anderson}}(\omega) = \frac{1}{\omega - \epsilon_0 - \Delta(\omega) - [\Sigma(\omega) - \Sigma_{\text{dc}}(\omega)]}$$

DFT+DMFT scheme



DFT+DMFT scheme

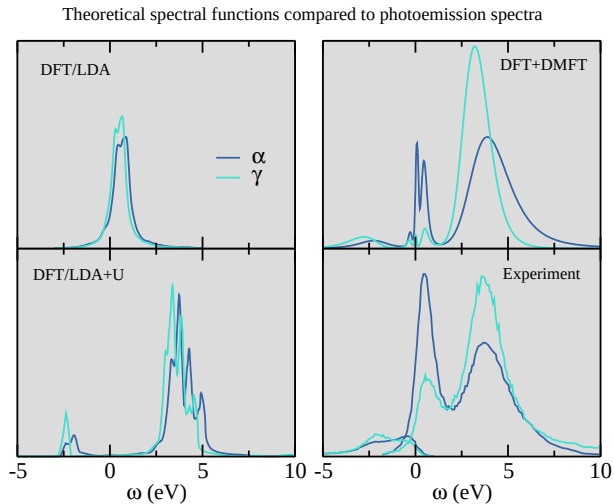


More generally, DFT+DMFT can be expressed as functional of the local Green's function and the electronic density $\Rightarrow$ Internal and free energies can be computed.

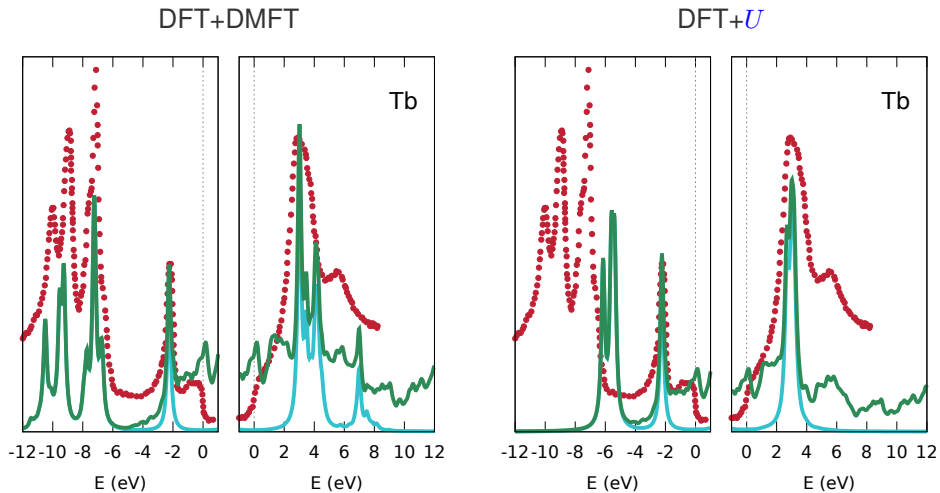
A functional of the density and the local Green's function

Theory	DFT	Green's fct Functional Theory
Interaction functional Interaction potential Equivalent system	$E_{\text{Ha+xc}}[n(\mathbf{r})]$ $v_{\text{Ha+xc}}$ non interacting system with a effective potential	$\widehat{\Phi}_U[G]$ Self energy Σ non interacting system with a frequency dependent potential (self energy)
Approximation Reference system	LDA Homogeneous electron gas	DMFT Anderson Impurity model

Spectral functions of cerium

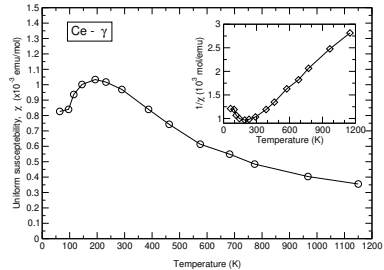
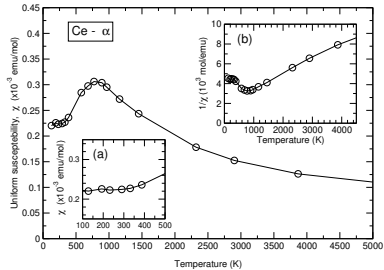


Spectral functions of Tb



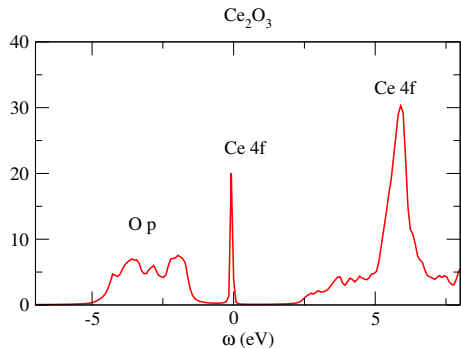
L. M. Locht et al, Phys. Rev. B 94, 085137 (2016)

Magnetic susceptibility



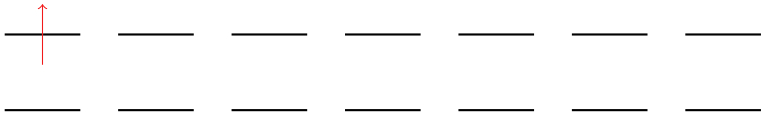
S. V. Streltsov, E. Gull, A. O. Shorikov, M. Troyer, V. I. Anisimov, and P. Werner, Phys. Rev. B 85, 195109 (2012)

Spectral function in Ce_2O_3



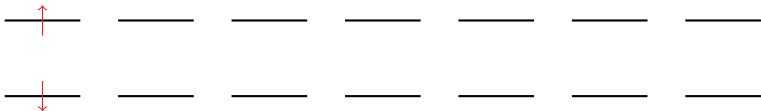
B. Amadon JPCM (2012)

DFT+DMFT can describe paramagnetic insulators



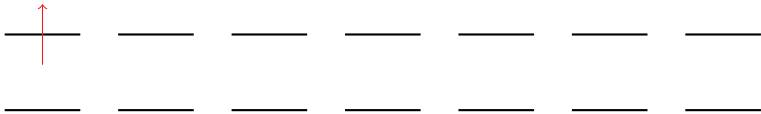
DFT+U: a configuration

DFT+DMFT can describe paramagnetic insulators



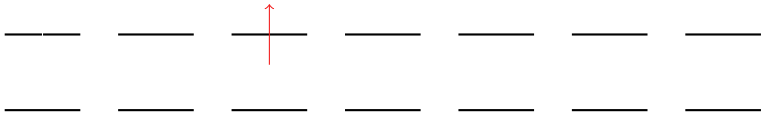
DFT+U: an instable configuration

DFT+DMFT can describe paramagnetic insulators



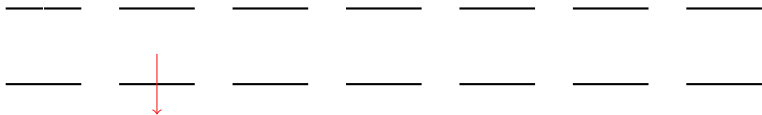
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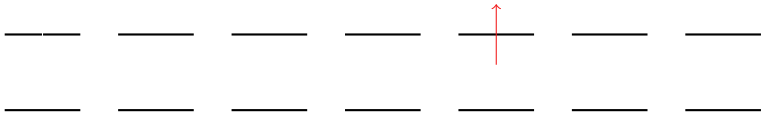
DFT+U: another configuration

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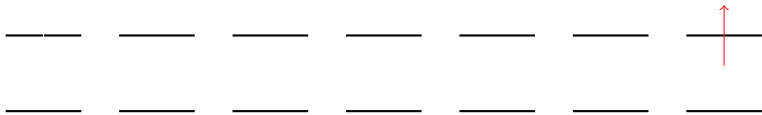
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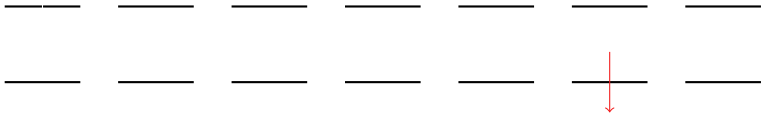
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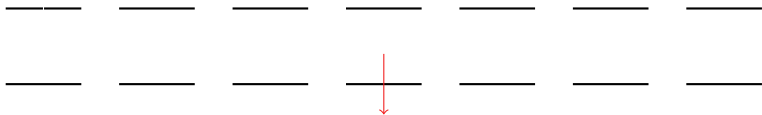
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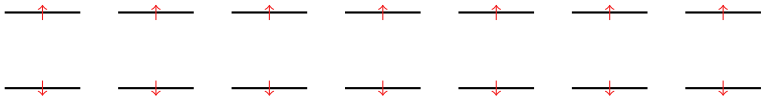
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DFT+U: another configuration

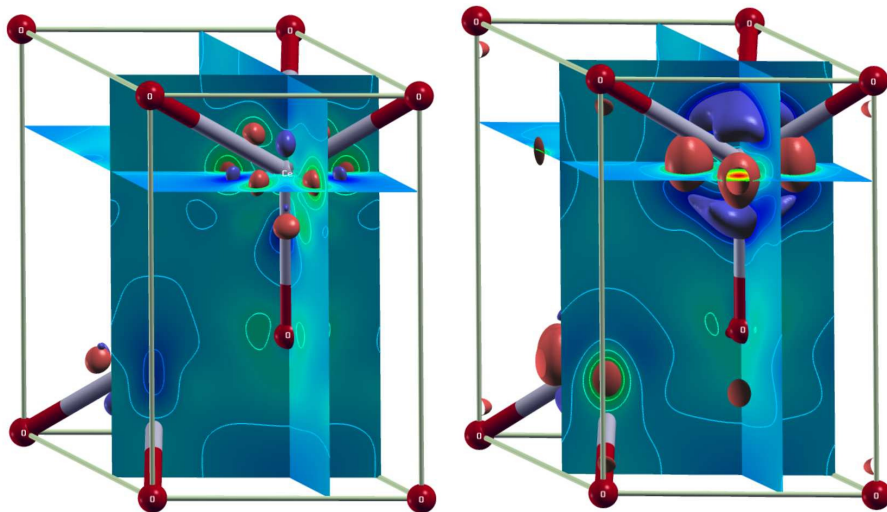
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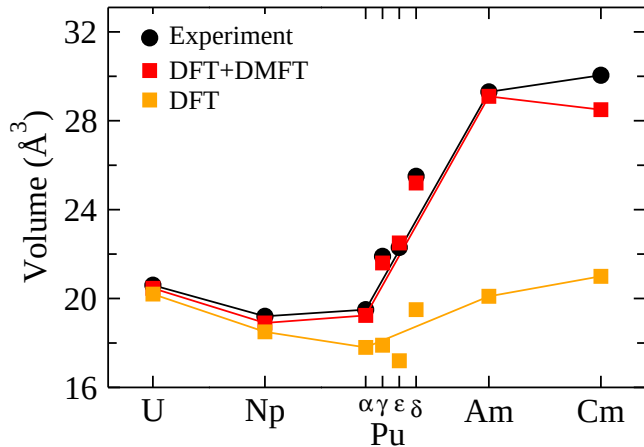
DFT+DMFT: More physical description

DFT+DMFT: Better description of magnetism

Electronic density in Ce_2O_3



Actinides in DMFT



B. Amadon PRB (2016)

Conclusion

- Strong correlations
 - Localized orbitals induce strong electronic Coulomb interaction.
 - This repulsion can induce strong localization of electrons and creates Mott insulator.
 - Hubbard bands and Kondo effects are signatures of strong interaction.
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- Perspectives
 - Include non local interactions and correlations: Cluster-DMFT
 - Treat also weak correlation effects and more ab-initio (no parameters): GW+DMFT